

Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context

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Abstract

A stored description has a rate, paid by the decoder, and a conditional content, paid by any party that keeps it while holding side information, and the two are set by different parts of the encoding configuration. An encoder observes a jointly Gaussian pair, a described variable and a correlated context, and describes the first for a decoder that sees the description alone. A third party holds side information, a noisy copy of the context, and never decodes. The minimal conditional content is Steinberg's common-reconstruction functional on the augmented source, and for jointly Gaussian sources it has a closed form, $L(D) = \frac{1}{2} \log_2 g^*$ with g^* the larger root of an explicit quadratic, which is the largest generalized eigenvalue of a matrix pencil and so extends to vector context. The region of jointly achievable rate and conditional content is exact, with a Pareto frontier traced by two coupled noise fractions. Rate and conditional content strictly trade off exactly when the context is partially informative, and the tradeoff persists when the side information is noiseless. The conditional content is not determined by the joint law of the described variable and the side information, so the encoder's access to the clean context enters the function. The rate-content frontier of the doubly symmetric binary case is computed explicitly, and coding theorems, including a direct proof for the Gaussian source, give the conditional content its operational meaning as the least normalized conditional entropy of the stored index, the rate at which a party holding the side information can re-encode it.

Index Terms

Common reconstruction, conditional mutual information, Gaussian sources, rate-distortion theory, side information.

I. INTRODUCTION

A stored description of data can carry more than one cost. The decoder that reconstructs from it pays its rate. A party that keeps the description and holds side information pays something else, the conditional entropy of the description given what it holds. That party can store or forward the description with a prefix-free code whose expected length is within one bit of this conditional entropy and no shorter, so it is the least rate at which the description can be re-encoded for that party. This paper proves, in the simplest setting where the question can be answered exactly, that these two costs are set by different parts of the encoding configuration and cannot in general be minimized together. Three facts carry the result. The second cost is not determined by the joint law of the reproduced variable and the side information, because it depends on what the encoder observes (Corollary 21). The two costs are minimized by different descriptions exactly when the encoder's extra observation is partially informative, and the gap between the two optima grows without bound as the side information becomes clean (Theorem 14(b), Corollary 19). And for jointly Gaussian sources the whole picture is explicit, the minimal conditional content in closed form and the rate-content region with its frontier. The doubly symmetric binary source is solved through a tilt equation.

A concrete instance is a two-tier data store (Fig. 1). A serving tier keeps the compressed records and answers queries from the descriptions alone; it pays the rate. An archival tier keeps a copy of every description together with metadata correlated with the records, which the encoder saw exactly and the archive holds only approximately, and it stores each description compressed against that metadata; it pays the conditional content. One encoder, which sees the record and its exact context, produces one description for both tiers, and the question is how cheap it can make the description for one tier without making it expensive for the other. The per-record costs in this paper

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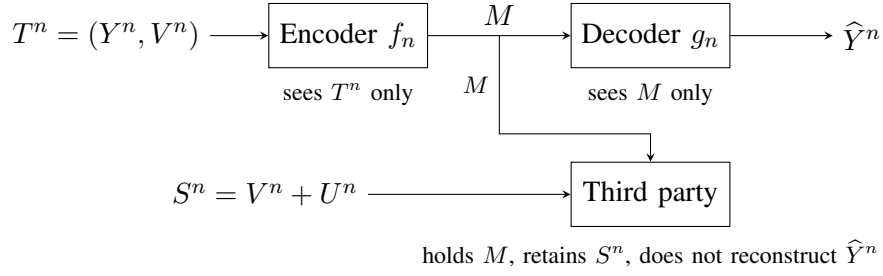


Fig. 1. The system. The encoder observes the source alone and stores the description M at rate R_n ; the decoder reconstructs from M alone; the third party retains the side information S^n and holds the stored index M ; it never reconstructs. The figure shows the Gaussian instantiation, where the encoder observes $T = (Y, V)$, $S = V + U$ with U independent of T , and the reproduction is $\hat{Y}^n$, an estimate of Y^n alone; the general discrete theorem reproduces the full letter.

are small numbers of bits, but such stores are large. At the scale of a trillion records, a size vector indexes now reach, every bit per record and coordinate is 125 GB, so the misalignment excesses of Section V correspond to at most about 14 GB of rate and 10 GB of content in a moderate regime, and to about 190 GB of rate near the clean-context boundary. This is arithmetic on the scale of the store, not a claim that any particular store follows the Gaussian model.

The setting is as follows. An encoder observes a jointly Gaussian pair (Y, V) : Y is the variable the decoder must reproduce, and V is a correlated variable, which we call the context ($\rho = \mathbb{E}[YV]$ after normalization). The encoder emits a description under the distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \leq D$, and the decoder reconstructs from the description alone. A third party holds $S = V + U$, a noisy observation of the context with noise variance τ^2 . The quantity of interest is $L(D) = \min I(X; \hat{X} | S)$ over test channels satisfying the distortion constraint and the Markov chain $\hat{X} - X - S$, which we call the minimal conditional content. By the operational theorems of Section II and, for the Gaussian source, Appendix A, it is the least asymptotically achievable value of the normalized conditional entropy $H(M | S^n)/n$ of the stored index M among codes meeting the distortion constraint. The same quantity has a thermodynamic reading, stated once in Remark 8 with the assumptions it needs; nothing else in the paper depends on it.

The single-letter functional itself is not new. With the augmented source $X = (Y, V)$ and a distortion that ignores the V coordinate, $\min I(X; \hat{X} | S)$ is exactly Steinberg's common-reconstruction functional [12], and Remark 8 of Lapidoth, Malär, and Wigger [13] notes precisely this augmentation device: an encoder observing an additional sequence correlated with the side information is handled by enlarging the source alphabet and letting the distortion ignore the extra coordinate. The region of jointly achievable rate and conditional content is likewise known in single-letter form, as the rate–distortion–equivocation region of secure source coding with side information at the eavesdropper [17, Theorem 3], since the conditional content is the leakage of the description to the party holding S (Related work). What this paper adds is the solution of the augmented Gaussian problem, and two contributions are primary. First, the closed form: we evaluate the augmented Gaussian functional exactly, $L(D) = \frac{1}{2} \log_2 g^*$ with g^* the larger root of an explicit quadratic together with the achieving channel (Theorem 14(a)), after Lemma 10 reduces the problem, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source, to the pair (ρ, τ^2) . The known Gaussian evaluations do not cover this one. Steinberg's Gaussian example [12] and the Gaussian theorem of [13] take a scalar source with side information $X + U$ and an encoder that observes the source alone; the augmentation remark of [13] is stated for finite alphabets; and the vector Gaussian common-reconstruction function of Lu and Xu [25] covers the augmented source only with the context penalized, under a hypothesis that excludes the unpenalized limit evaluated here (Appendix C). Second, the exact region of jointly achievable rate and conditional content. Its Pareto frontier is traced by a stationarity system in two coupled noise fractions, one for each coordinate, and the two coordinates are minimized by different test channels, so that rate and content strictly trade off, exactly when $0 < \rho^2 < 1$ within the noisy model $\tau^2 > 0$; the tradeoff persists at the clean-context boundary (Theorem 14(b) and Corollary 19). We call this separation of the two optima misalignment. The size of the misalignment is a property of the regime. It grows without bound toward the clean-context boundary, where the content-optimal description pins the reproduction to the context and its rate diverges, for instance by 1.54 bits at $(\rho^2, \tau^2, D) = (0.5, 10^{-3}, 0.5)$. Over a moderate parameter box it stays below 0.12 bits of rate and 0.08 bits of

content (Section V). One consequence of the closed form is that $L(D)$ is not determined by the joint law of (Y, S) (Corollary 21): two sources whose (Y, S) laws agree up to invertible scaling can have different minimal conditional content, because the encoder's access to the clean context V , which neither the decoder nor the third party shares, enters the function itself.

The closed form has a second description, which explains where the quadratic comes from. The minimal conditional content is the largest generalized eigenvalue of a matrix pencil formed from the covariance of the encoder's observation, its conditional covariance given the side information, and the distortion level (Proposition 24), and the same formula gives the vector-context function explicitly. A companion manuscript [39] treats a described variable of dimension greater than one, where the eigenvector of that pencil, the direction the description reads, turns with the distortion budget.

The remaining results are extensions and classical limits. The closed form recovers Shannon's rate-distortion function at $\rho = 0$ [1], Gray's conditional function as $\tau^2 \rightarrow 0$ [4], and Steinberg's scalar Gaussian formula at the induced correlation as $\rho^2 \rightarrow 1$ (Corollary 15), and marginalization is strictly suboptimal (Corollary 16). Remark 22 gives the attainment condition for the conditional-variance bound. The frontier extends to vector context (Section VI); the rate-content frontier of a doubly symmetric binary source with a binary symmetric context channel is traced through an elementary tilt equation (Section VII); and the operational region theorem is proved directly for the Gaussian source (Appendix A).

Related work

Conditional rate-distortion theory is due to Gray [3], [4]: both terminals observe the conditioning variable, and $R_{Y|S}(D)$ prices the description under shared context. Here no terminal observes S , and Gray's function appears only as the clean-context limit $\tau^2 \rightarrow 0$ (Section IV). Wyner and Ziv [5] place the side information at the decoder, where it participates in reconstruction; our decoder holds nothing but the description, and Wyner-Ziv binning enters the operational theorem only as a proof device.

Steinberg [12] introduced the common-reconstruction constraint, under which the decoder holds side information but the encoder must be able to reproduce the decoder's output; the resulting functional $\min I(X; \hat{X} | S)$ under $\hat{X} - X - S$ is the single-letter object of this paper, and Steinberg solved its scalar Gaussian case. Lapidot, Malär, and Wigger [13] relaxed exact to approximate reproduction and solved the scalar Gaussian problem with side information $X + U$; their Remark 8, stated for finite alphabets, incorporates an encoder observing an additional correlated sequence by augmenting the source and letting the distortion ignore the extra coordinate. Our problem is exactly that augmented problem, with $X = (Y, V)$ and distortion on Y alone. What is inherited is the functional and the augmentation device; what is new, to our knowledge, is its evaluation and the joint region of the two coordinates in place of the rate under a common-reconstruction constraint alone. The encoder-side-information common-reconstruction model of Lu *et al.* [23], [24] is this paper's formulation for the binary source: under the relabeling $X \leftrightarrow Y$, $S_1 \leftrightarrow V$, $S_2 \leftrightarrow S$, $W \leftrightarrow \hat{Y}$ their functional is the binary conditional-content function. Their conditioned quantity is a transmitted rate with a Wyner-Ziv saving, while here the transmitted rate is the unconditional R and the conditioned quantity is a separate cost, so their single-coordinate problem does not constrain R ; the binary contribution here is the joint frontier (Appendix C).

Side information held by the encoder alone is classically inert: when it does not enter the distortion measure, it leaves the rate-distortion function unchanged [2], and Kaspi [10] analyzed the configurations in which the encoder is informed of the side information. Here the encoder-held context V matters because the second coordinate is conditional on S , with which V is correlated, and access to V strictly lowers it for every $\rho \neq 0$ (Corollary 16 for $\tau^2 > 0$; Proposition 11 at the clean-context boundary). Heegard and Berger [11] serve decoders with and without side information from one description, and Ahmadi, Tandon, Simeone, and Poor [14] imposed the common-reconstruction constraint on the Heegard-Berger and cascade problems; Section VII distinguishes the binary frontier from their calculations.

The closest Gaussian counterpart is Lu and Xu's vector Gaussian Heegard-Berger problem with common reconstructions [25]. Its closed form keeps the context penalized by its own distortion constraint, and Appendix C shows that it lies strictly above $L(D)$ on its admissible range and does not continue to the unpenalized limit evaluated here. The pencil characterization of Proposition 24 parameterizes a Gaussian description by its error covariance, a standard device in vector Gaussian source coding [40].

The conditional coordinate is the information leakage of secure source coding. For a deterministic encoder, $H(X^n | M, S^n) = H(X^n | S^n) - H(M | S^n)$, so $L_n = H(M | S^n)/n = I(X^n; M | S^n)/n$ is the leakage of the description to a party holding S^n beyond what S^n reveals, and minimizing it is the same as maximizing that party's equivocation about the source. Villard and Piantanida [17, Theorem 3] characterized the rate–distortion–equivocation region of secure lossy source coding with side information at the eavesdropper; without decoder side information, and with the source alphabet augmented to (Y, V) , their region is the region of Theorem 6, which we prove again only to record the finite-blocklength constants that Appendix A uses. Ekrem and Ulukus [18] obtained the minimum leakage of a vector Gaussian source with side information at both receivers when the rate is unconstrained, under a matrix distortion on the whole source, and reported that they could not solve the weighted problem that traces the joint rate–leakage region. The present paper evaluates that joint region in closed form for the informed-encoder source without decoder side information, where the context is unpenalized; their matrix constraint on the whole source does not cover this case, and it is the encoder's access to the context that makes the two costs disagree. Yamamoto priced distortion against source equivocation in the Shannon cipher system [15] and, earlier [16], let the encoder observe a source together with an additional output to be kept secret; Schieler and Cuff [19] developed a rate–distortion theory for secrecy systems. The storage reading of the same quantity, the least rate at which the party holding the side information can re-encode the description, is the one this paper emphasizes. Closer in form are the side-information-privacy problem of Tandon, Sankar, and Poor [20] and the privacy funnel of Makhdoumi, Salamati, Fawaz, and Médard [21], which price the leakage $I(\hat{X}; S)$ of the description about the side information. Under the Markov chain $I(X; \hat{X} | S) = I(X; \hat{X}) - I(\hat{X}; S)$, so the present problem rewards the leakage that the privacy line penalizes: the two share their coordinates with opposite signs.

The relation to the information bottleneck [28] is exact. For any channel with $\hat{Y} = (Y, V) - S$, writing $T = (Y, V)$, the weighted objective of Section V collapses by the Markov chain:

$$J_\alpha := \alpha I(T; \hat{Y}) + (1 - \alpha) I(T; \hat{Y} | S) = I(T; \hat{Y}) - (1 - \alpha) I(S; \hat{Y}), \quad (1)$$

an information-bottleneck Lagrangian with tradeoff parameter $1 - \alpha$, augmented by the hard distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \leq D$. Gaussian IB [27] establishes the optimality of Gaussian linear-noise representations and the eigenmode structure of the unconstrained Lagrangian, and Meidlinger, Winkelbauer, and Matz [26] connected it to mean-square rate–distortion quantization on the rate coordinate; the hard Y -distortion constraint binds a reconstruction error rather than a mutual information, and the two-noise-fraction frontier is what it adds.

Finally, [37] is a preliminary manuscript by the author, archived at doi:10.5281/zenodo.21776291, where the discrete operational region theorem, its single-letterization, and the scalar Gaussian corner first appeared; everything from Section III onward is developed here, with the pair-sufficiency reduction credited to Armstrong's source-side sufficiency [22] as a distortion-preserving specialization.

Why the evaluation is not routine

The functional is known, so it is fair to ask whether its Gaussian evaluation is routine. Four features keep it from being so. First, the distortion constrains Y alone, yet the optimal channel loads on the context, with coefficient $b = (g^* - 1)\rho/(g^*k)$ nonzero for every $\rho \neq 0$ (Theorem 14(a)). Under the Markov chain the content is $I(T; \hat{Y}) - I(\hat{Y}; S)$, so loading the description on the context, which the side information observes, lowers its content even though it does nothing for the distortion. The reverse water-filling on Y that solves the classical and Gray problems therefore does not apply, and the optimization runs over the joint regression of the reproduction on (Y, V) against the conditional covariance of (Y, V) given S . Second, the region is not traced by one water level. Its frontier is the solution of a coupled stationarity system in two noise fractions, and the existence and uniqueness of that solution at every weight is a separate result (Proposition 18), proved through a convexity lemma in transformed coordinates (Lemma 17). Third, at the clean-context boundary $\tau^2 = 0$ the conditional covariance of (Y, V) given S is singular, the exhaustion lemma behind the converse (Lemma 12) no longer applies, and the persistence of the tradeoff there needs its own argument (Propositions 11 and 20). Fourth, the operational meaning of the second coordinate is the conditional entropy of a discrete index given a continuous side information, which does not follow from the finite-alphabet theorem by a standard limit. Appendix A proves the Gaussian operational theorem directly through a quantized conditioner, and the same obstruction is why the finite-blocklength dispersion of the content coordinate remains open (Section VIII).

II. PROBLEM FORMULATION AND THE OPERATIONAL THEOREM

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. The model has one source and three parties, shown in Fig. 1. Let $(X, S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. The theorem of this section is for a general discrete source letter X ; the Gaussian specialization, in which X becomes the pair $T = (Y, V)$ and $S = V + U$, is fixed once, at the start of Section III. The encoder observes X^n alone and emits the description M . The decoder observes M alone and outputs the reproduction; the general discrete theorem below reproduces the full letter, $\hat{X}^n$, while in the Gaussian sections the reproduction is a scalar estimate $\hat{Y}$ of Y alone. The third party of Section I retains S^n and holds the stored index M ; it never decodes.

A distortion measure is a bounded function

$$d : \mathcal{X} \times \hat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \quad (2)$$

An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \rightarrow \mathcal{M}_n, \quad g_n : \mathcal{M}_n \rightarrow \hat{\mathcal{X}}^n. \quad (3)$$

We write $M = f_n(X^n)$ and $\hat{X}^n = g_n(M)$. Randomized encoders do not enlarge the region below: the converse allows them and the achievability does not need them.

The three operational quantities are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad (4)$$

$$D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} d(X_i, \hat{X}_i), \quad (5)$$

$$L_n = \frac{1}{n} H(M | S^n). \quad (6)$$

The dimensionless quantity L_n is the uncertainty about the stored index that remains given the retained side information, the equivocation of the index at the third party in the sense of the secrecy literature (Section I). We call it the *conditional content* of the stored description, because here it is a cost to be made small rather than a protection to be made large. Every statement and proof of the paper treats L_n as the coding quantity (6), in bits per source symbol.

Definition 1. A triple (R, L, D) is achievable if there is a sequence of observation codes such that

$$\limsup_{n \rightarrow \infty} R_n \leq R, \quad \limsup_{n \rightarrow \infty} L_n \leq L, \quad \limsup_{n \rightarrow \infty} D_n \leq D. \quad (7)$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}(D)$.

For a test channel $p(\hat{x} | x)$, the Markov chain $\hat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}(D) = \left\{ p(\hat{x} | x) : \mathbb{E} d(X, \hat{X}) \leq D \right\}. \quad (8)$$

The two single-letter functions of the paper are the classical rate–distortion function and the content endpoint,

$$R_{\min}(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \hat{X}), \quad L(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \hat{X} | S). \quad (9)$$

Both minima exist by compactness of $\mathcal{T}(D)$ and continuity of the two coordinates. The coding theorem below gives both functions their operational meaning. The following definition fixes the two notational scopes of the paper.

Definition 2. General letters, used for the discrete theorem of this section: source $(X, S) \sim p_{XS}$, i.i.d. copies (X^n, S^n) , test channel $p(\hat{x} | x)$ with the Markov chain $\hat{X} - X - S$, constraint set $\mathcal{T}(D)$, and the two functions of (9). Gaussian letters, used from Section III on: the source letter is the jointly Gaussian pair $T = (Y, V)$ with $\text{Var } Y = \text{Var } V = 1$ and $\mathbb{E}[YV] = \rho$; the context copy is generated as $S = V + U$ with $U \sim \mathcal{N}(0, \tau^2)$ independent of everything else; the test channel is $p(\hat{y} | t)$, the reproduction $\hat{Y}$ does not see S , and the Markov chain is $\hat{Y} - T - S$; the constraint is $\mathbb{E}(Y - \hat{Y})^2 \leq D$; and

$$R_{\min}(D) = \min I(T; \hat{Y}), \quad L(D) = \min I(T; \hat{Y} | S),$$

both minima over test channels meeting the constraint.

Lemma 3. For fixed p_{XS} , both coordinates $I(X; \hat{X})$ and $I(X; \hat{X} | S)$ are convex functions of the test channel $q(\hat{x} | x)$.

Proof. Convexity of $I(X; \hat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\hat{X} - X - S$ gives $p(\hat{x} | x, s) = q(\hat{x} | x)$, so

$$I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q), \quad (10)$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x | s)$ and the common channel q ; each term is convex in q , hence so is the sum. We note that the decomposition $I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X})$ does *not* yield this conclusion: $I(S; \hat{X})$ is itself convex in q (the induced channel $p(\hat{x} | s) = \sum_x p(x | s)q(\hat{x} | x)$ is linear in q), so the difference is a priori indefinite. $\square$

Remark 4. Lemma 3 holds for standard Borel alphabets, with the sums over s replaced by integrals against the law of S , the conditional laws furnished by regular conditional probabilities, and conditional mutual information defined as the expected divergence of the conditional laws [7]: the proof uses only the joint convexity of relative entropy [41] and the linearity of the induced output law in the channel, and the convexity holds with values in $[0, \infty]$. Theorem 14(b) and Theorem 29 use this form.

Lemma 5. For a finite-alphabet source (X, S) and bounded distortion d , define

$$\mathcal{R}(D) := \bigcup_{q \in \mathcal{T}(D)} \{(R, L) : R \geq I_q(X; \hat{X}), L \geq I_q(X; \hat{X} | S)\}.$$

Then $\mathcal{R}(D)$ is closed and nondecreasing in D , and it is right-continuous:

$$\mathcal{R}(D) = \bigcap_{D' > D} \mathcal{R}(D').$$

Equivalently, if $(R_j, L_j) \in \mathcal{R}(D_j)$ with $(R_j, L_j) \rightarrow (R, L)$ and $\limsup_j D_j \leq D$, then $(R, L) \in \mathcal{R}(D)$.

Proof. Monotonicity is the nesting of the constraint sets $\mathcal{T}(D)$. For the sequential form, pick $q_j \in \mathcal{T}(D_j)$ with $R_j \geq I_{q_j}(X; \hat{X})$ and $L_j \geq I_{q_j}(X; \hat{X} | S)$. The channels live in the compact simplex of conditional distributions, so a subsequence converges to some q ; the distortion $\mathbb{E}_q d(X, \hat{X})$ is continuous in the channel, so $\mathbb{E}_q d = \lim_j \mathbb{E}_{q_j} d \leq \limsup_j D_j \leq D$ and $q \in \mathcal{T}(D)$. Both mutual-information coordinates are continuous in the channel on finite alphabets, so $I_q(X; \hat{X}) = \lim_j I_{q_j}(X; \hat{X}) \leq R$ and likewise $I_q(X; \hat{X} | S) \leq L$: the limit (R, L) lies in q 's quadrant, hence in $\mathcal{R}(D)$. Closedness is the sequential form with $D_j \equiv D$. For right-continuity, the inclusion $\mathcal{R}(D) \subseteq \bigcap_{D' > D} \mathcal{R}(D')$ is monotonicity, and conversely a point of the intersection is a constant sequence with $(R_j, L_j) \equiv (R, L) \in \mathcal{R}(D'_j)$ along any $D'_j \downarrow D$, so it lies in $\mathcal{R}(D)$ by the sequential form. $\square$

Outline of the coding theorem: The reconstruction decoder never sees S^n . The rate comes from ordinary covering, and reconstruction is from the full index M . A *virtual* bin index $B = b(M)$, which no party transmits or stores, and a Fano bound at a *hypothetical* decoder holding (B, S^n) serve only to bound $H(M | S^n)$; an expurgation step then selects one deterministic code meeting both requirements.

Theorem 6. For a discrete memoryless source (X, S) and bounded distortion d ,

$$\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}(D)} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}. \quad (11)$$

By Lemma 3 both coordinates are convex in the common test channel, so the union is already convex, and it is closed and right-continuous in the distortion level (Lemma 5); consequently no time sharing is required, and every Pareto-optimal point of the region is attained by a single test channel.

The region is [17, Theorem 3] without decoder side information, read through the identity $H(X^n | M, S^n) = H(X^n | S^n) - H(M | S^n)$; the proof below is included for the finite-blocklength constants of Lemma 7.

Proof. We first prove the converse. Let a code produce M and $\hat{X}^n$. Since $\hat{X}^n$ is a function of M ,

$$\begin{aligned} \log |\mathcal{M}_n| &\geq H(M) \geq I(X^n; M) \geq I(X^n; \hat{X}^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i), \end{aligned} \quad (12)$$

where the last step follows from memorylessness and conditioning reducing entropy (the standard single-letterization, as in [5]). Likewise,

$$\begin{aligned} H(M | S^n) &\geq I(X^n; M | S^n) \geq I(X^n; \hat{X}^n | S^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i | S_i). \end{aligned} \quad (13)$$

Each induced per-letter channel $p(\hat{x}_i | x_i)$ satisfies the Markov constraint $\hat{X}_i - X_i - S_i$: $\hat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} | x) = \frac{1}{n} \sum_{i=1}^n p(\hat{X}_i = \hat{x} | X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s | x)$ is common across i . Its distortion is D_n , so $\bar{q} \in \mathcal{T}(D_n)$, and by Lemma 3,

$$R_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i) \geq I_{\bar{q}}(X; \hat{X}), \quad L_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i | S_i) \geq I_{\bar{q}}(X; \hat{X} | S),$$

which places (R_n, L_n) in the (already convex and closed) union of (11) at distortion D_n . The right side of (11) is the set $\mathcal{R}(D_n)$ of Lemma 5, so, for an achievable triple with $\limsup D_n \leq D$, the sequential form of that lemma places every limit point of (R_n, L_n) , a bounded sequence by Definition 1, and with it (R, L) , in the right side of (11) at D .

For achievability, fix $p(\hat{x} | x) \in \mathcal{T}(D)$ and $\epsilon > 0$. Generate a random codebook $\mathcal{C}$ of $2^{n(I(X; \hat{X}) + 2\epsilon)}$ independent $\hat{x}^n$ codewords drawn from the $\hat{X}$ -marginal, and, independently, a random binning b assigning each codeword index uniformly to one of $2^{n(I(X; \hat{X} | S) + 3\epsilon)}$ bins. The covering encoder selects the smallest index among the codewords jointly (strongly) typical with x^n , or index 1 if there is none, and stores that index M ; the decoder reconstructs the codeword from M alone. There are two failure events. First, E_{cov} : no jointly typical codeword exists; $\Pr(E_{\text{cov}}) \rightarrow 0$, averaged over $(X^n, \mathcal{C})$, by the covering lemma, and on E_{cov}^c the distortion is at most $D + \epsilon$ by typicality. Second, E_{bin} : the side-information proof device fails. Because $\hat{X} - X - S$,

$$I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X}), \quad (14)$$

so each bin holds $2^{n(I(S; \hat{X}) - \epsilon)}$ codewords on average, a per-bin codeword rate ϵ below the packing threshold $I(S; \hat{X})$, and the Wyner–Ziv packing argument [5] (the Markov lemma for the typicality of the selected codeword with S^n , and the packing bound for wrong codewords in the bin) gives a decoder that recovers M from (B, S^n) , $B = b(M)$, with $\Pr(E_{\text{bin}}) \rightarrow 0$ averaged over $(X^n, S^n, \mathcal{C}, b)$: it outputs the unique codeword in bin B jointly typical with S^n and declares an error otherwise. If $I(S; \hat{X}) = 0$, so that $I(X; \hat{X} | S) = I(X; \hat{X})$, the number of bins exceeds the number of codewords and the bin index determines M with high probability; the conclusion is unchanged. This decoder is a proof device; the actual decoder of the code still receives no side information.

Selection of one deterministic code. Let $\epsilon_n := \mathbb{E}_{\mathcal{C}, b} [\Pr(E_{\text{cov}} | \mathcal{C}) + \Pr(E_{\text{bin}} | \mathcal{C}, b)] \rightarrow 0$. By Markov's inequality there is a deterministic pair $(\mathcal{C}, b)$ with $\Pr(E_{\text{cov}}) + \Pr(E_{\text{bin}}) \leq 2\epsilon_n$. Fix it. Its distortion satisfies $D_n \leq D + \epsilon + 2\epsilon_n d_{\text{max}}$, and by Fano's inequality applied to the proof device on this fixed code, $H(M | B, S^n) \leq 1 + 2\epsilon_n \log |\mathcal{M}_n| = 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$, so

$$\begin{aligned} H(M | S^n) &\leq H(B) + H(M | B, S^n) \\ &\leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon) = n(I(X; \hat{X} | S) + 3\epsilon + o(1)). \end{aligned} \quad (15)$$

The selected deterministic code thus satisfies the distortion and the conditional-entropy requirements simultaneously: the stored index has ordinary description rate $I(X; \hat{X}) + 2\epsilon$, distortion at most $D + \epsilon + o(1)$, and conditional content at most $I(X; \hat{X} | S) + 3\epsilon + o(1)$. Letting $\epsilon \downarrow 0$ along a diagonal sequence of blocklengths exhibits the corner $(I(X; \hat{X}), I(X; \hat{X} | S))$ at distortion D ; since the union in (11) is already convex and closed (Lemmas 3 and 5), this exhibits the entire region. $\square$

The achievability construction delivers finite-blocklength guarantees that Appendix A invokes on a quantized surrogate; we record them as a lemma.

Lemma 7. *Let (X, S) be a discrete memoryless source with finite alphabets, d a distortion with maximum $d_{\max}$, $p(\hat{x} | x)$ a test channel with $\hat{X} - X - S$ and finite reproduction alphabet, and $\epsilon > 0$. There is a sequence $\epsilon_n \rightarrow 0$, depending on the joint law, the channel, ϵ , and, through the typicality tolerance that keeps the covered distortion within ϵ , on $d_{\max}$ and the alphabet sizes, such that for all sufficiently large n there is a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X;\hat{X})+2\epsilon)}$, distortion $D_n \leq \mathbb{E} d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\max}$, and*

$$H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon).$$

Proof. This is the selected code of the achievability part of the proof of Theorem 6, with the constants displayed at (15); ϵ_n is the averaged failure probability defined there. $\square$

The binning step in the proof has a simple interpretation. The full codeword index is required by the decoder. Given the side information, the codeword is identified up to a residual set of asymptotic size $2^{nI(X;\hat{X}|S)}$; only that residual uncertainty remains in the stored index.

Remark 8. *This is the paper's only physical claim; everywhere else L_n and $L(D)$ are the coding quantities (6) and (9). Take the reset model of stochastic thermodynamics with side information [34], [36]. The third party holds the record M and the classical side information S^n , which it reads without disturbance, and no other record correlated with M ; neither X^n nor $\hat{Y}^n$ is available to it (with X^n available, M could be uncomputed at no cost [30]). It knows the joint law of (M, S^n) , may drive M by any protocol in contact with a bath at temperature T_b whose driving may depend on S^n but not on M , and only the mean work is counted. The memory is Landauer's ideal logical memory: symmetric logical states with no free-energy difference between them [35], a protocol cyclic in its Hamiltonian, and the side information left unchanged. In that model the generalized second law with correlations [34], [36] bounds the average work from below by $k_B T_b \ln 2 H(M | S^n)$, Landauer's bound for arbitrary distributions [29], [33], and running for each value s of S^n the optimal erasure of $p(m | s)$ approaches it, though not in finite time [33]. Hence the infimum of the average work over protocols is*

$$W_n^* = k_B T_b \ln 2 H(M | S^n) = n k_B T_b \ln 2 L_n \quad (16)$$

for every code. A reader of S^n at finite resolution pays at least as much, with an excess that vanishes as the resolution is refined (Appendix A). By Theorem 6, and by Theorem 29 for the Gaussian source, $L(D)$ is therefore the infimum of the asymptotic average erasure work per source symbol, in units of $k_B T_b \ln 2$, over codes meeting the distortion constraint. The claim is for average work; the single-shot setting prices the same reset by smooth conditional entropies [31], [32], and we claim nothing there. It counts only the logically irreversible reset of M , and not the work of acquiring X^n , driving a channel, correcting errors, or finishing in finite time, nor the codebook and wiring, which are not erased in each cycle [30], [29].

As an information quantity, the content endpoint $L(D)$ is Steinberg's common-reconstruction rate-distortion function [12], but the roles differ. There S is decoder side information and $I(X; \hat{X} | S)$ is the transmitted rate; here S is held by a third party that never serves the decoder, the transmitted rate is the undiscounted $I(X; \hat{X})$, and $I(X; \hat{X} | S)$ is a second cost. The object of the paper is therefore the joint region rather than either endpoint.

Remark 9. *Theorem 6 is for finite alphabets and bounded distortion. For the jointly Gaussian source with quadratic distortion the operational statement is Theorem 29 (Appendix A), and the Gaussian closed forms below, which evaluate the single-letter coordinates of (11), inherit their operational meaning through it.*

Table I collects the notation used from here on.

III. PAIR SUFFICIENCY

Let $X \sim \mathcal{N}(0, \Sigma_X)$ on $\mathbb{R}^d$, i.i.d. across source symbols. The described variable is the linear functional $Y = w^\top X$; the third party's copy is a second linear functional observed in noise, $S = \gamma^\top X + U$. Normalize $\text{Var}(w^\top X) = 1$ (rescaling D), write $V = \gamma^\top X / \sigma_\gamma$, $\sigma_\gamma^2 := \gamma^\top \Sigma_X \gamma$, the normalized context, with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$. Write $s = 1 + \tau^2$,

TABLE I
NOTATION.

Symbol	Meaning
$T = (Y, V)$	the encoder-observed pair, described variable Y and context V , normalized in Section III. X is the source letter of the discrete theorem (Section II) and, in the Gaussian sections, the ambient vector from which T is drawn
$S = V + U$	the side information, a noisy copy of the context; $U \sim \mathcal{N}(0, \tau^2)$ independent of X
$s = 1 + \tau^2$	the context-noise parameter, the variance of S after normalization (scalar side information; the vector-context section uses Σ_U , with side-information covariance $\Sigma_S = \Sigma_V + \Sigma_U$)
$\rho = \mathbb{E}[YV]$	correlation of the described variable and the context
D	distortion level, $\mathbb{E}(Y - \hat{Y})^2 \leq D$
M, R_n, L_n	stored index; rate $\frac{1}{n} \log \mathcal{M}_n $; conditional content $\frac{1}{n} H(M S^n)$
$R_{\min}(D), L(D)$	the two single-letter functions of (9)
$g^*; g_m, g_f, g_R$	the root giving $L(D) = \frac{1}{2} \log_2 g^*$ (Theorem 14); the roots compared with it: Steinberg's scalar formula on (Y, S) (Corollary 16), the conditional-variance bound (Remark 22), and Lu and Xu's largest admissible value (Appendix C)

the context-noise parameter (a scalar, not to be confused with the random variable S). Throughout, X , U , and all channel randomness (including any resampling randomness below) are *mutually* independent. This joint form of the independence assumption is essential: it is exactly what the S -kernel argument in the next proof uses.

From here on the Gaussian scope of Definition 2 is in force and is stated once. The source letter is the pair $T = (Y, V)$, drawn from the ambient vector X above, which enters only through T once Lemma 10 is proved; the distortion is $(y - \hat{y})^2$ on the Y coordinate, so the decoder is judged on Y alone while the encoder observes the pair; the reproduction is $\hat{Y}$; and all information quantities are written on $(T, \hat{Y}, S)$. Reproductions may be taken scalar without loss of optimality: replacing $\hat{X}$ by $\hat{Y} := w^\top \hat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot | S)$ (conditional data processing), and $\hat{Y} - X - S$ persists.

Standing assumptions and degeneracies: The Gaussian sections run under $0 < D < 1$, $|\rho| < 1$, and $\tau^2 > 0$; each boundary is treated where it arises. The cases $D \geq 1$ (where $R = L = 0$) and $D = 0$ (where no finite pair is achievable) are recorded with Theorem 29; the boundaries $\rho^2 \in \{0, 1\}$ are corners of Corollary 15, with the frontier collapse computed in Corollary 19; and the clean-context boundary $\tau^2 = 0$, where $\text{Cov}(T, S)$ is singular and Lemma 12 does not apply, is treated by Propositions 11 and 20. $\Sigma_T \succ 0$ is assumed, with linearly dependent components dropped where noted, and singular channel residuals are dispatched by Remark 13.

Lemma 10. *In the problem $L(D) = \min\{I(X; \hat{Y} | S) : \hat{Y} - X - S, \mathbb{E}(Y - \hat{Y})^2 \leq D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. Consequently $L(D)$ depends on $(\Sigma_X, w, \gamma, \tau^2)$, for the scalar linear functionals $Y = w^\top X$ and $V = \gamma^\top X / \sigma_\gamma$ of a source of arbitrary dimension d , only through (ρ, τ^2) .*

Proof. Given an admissible $p(\hat{y} | x)$, draw $\hat{Y}'$ from the conditional law $p(\hat{y} | T = t)$ using fresh randomness. (a) $(T, \hat{Y}') \stackrel{d}{=} (T, \hat{Y})$, so the distortion (a functional of $(Y, \hat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \hat{Y})$ and of $(X, \hat{Y}')$, so in both joints the conditional density of S given $(T, \hat{Y}) = (t, \hat{y})$ is $p_U(\cdot - v)$; hence $(T, \hat{Y}', S) \stackrel{d}{=} (T, \hat{Y}, S)$. (c) Given T , $\hat{Y}'$ is independent of (X, S) , so $I(X; \hat{Y}' | S) = I(T; \hat{Y}' | S) + I(X; \hat{Y}' | T, S) = I(T; \hat{Y} | S) \leq I(X; \hat{Y} | S)$, the last step because T is a function of X . (d) $\hat{Y}' - X - S$ holds since $\hat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U) . The same computation applied to a T -channel directly gives $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. $\square$

This reduction is the case $T = (Y, V)$ of Armstrong's source-side sufficiency for the information bottleneck [22], with the addition that the T -measurable distortion constraint is preserved, so it holds for the constrained functional and not only the unconstrained bottleneck.

IV. THE FUNCTION: SCALAR CONTEXT

One further reduction identifies the class the closed form must beat. (*Marginalized class = scalar corner.*) If the channel depends on X only through Y (that is, $\hat{Y} - Y - (X, S)$), then given (Y, S) , $\hat{Y}$ is independent of X , so $I(X; \hat{Y} | S) = I(Y; \hat{Y} | S)$, and the class optimum is exactly the scalar-corner value $\frac{1}{2} \log_2[(\sigma^2(1 - \rho_{YS}^2) +$

$\rho_{Y|S}^2 D)/D]$ of the pair (Y, S) , where $\sigma^2 = \text{Var } Y$ and $\rho_{Y|S}$ is the correlation of that pair, $\rho_{Y|S}^2 = \rho^2/s$ after the normalization of Section III [12]; the corner's converse applies to every $p(\hat{y} | y)$, discrete or not.

Proposition 11. *Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, $Y = X_1$, and $S = X_1 + X_2$ (so $\rho_{Y|S}^2 = \frac{1}{2}$ and S is X -measurable). For $0 < D < 1$:*

- (i) *the marginalized-class optimum is the scalar corner $L_{\text{marg}}(D) = \frac{1}{2} \log_2 \frac{1+D}{2D}$;*
- (ii) *the vector problem has the exact value*

$$L_{\text{vec}}(D) = \frac{1}{2} \log_2^+ \frac{1}{2D} = R_{Y|S}(D),$$

Gray's conditional rate-distortion function of (Y, S) [4] ($\text{Var}(Y | S) = \frac{1}{2}$), attained by the linear channel $\hat{Y} = (1-D)X_1 + DX_2 + N$, $N \sim \mathcal{N}(0, D(1-2D))$, for $D \leq \frac{1}{2}$, and by $\hat{Y} = S/2$ (zero conditional content) for $D \geq \frac{1}{2}$;

- (iii) *hence the strict gap $L_{\text{marg}}(D) - L_{\text{vec}}(D) = \frac{1}{2} \log_2(1+D) > 0$ on $(0, \frac{1}{2}]$, and equals the full marginalized value on $[\frac{1}{2}, 1)$ (where $L_{\text{vec}} = 0$; the absolute gap is decreasing there, but it is all of L_{marg}).*

Proof. (i) is the reduction above, evaluated at $\sigma^2 = 1$, $\rho_{Y|S}^2 = \frac{1}{2}$.

(ii) *Converse.* For any admissible $\hat{Y}$ (Markov $\hat{Y} - X - S$), $I(X; \hat{Y} | S) \geq I(Y; \hat{Y} | S)$, and the induced conditional channel $p(\hat{y} | y, s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y; \hat{Y} | S) \geq R_{Y|S}(D) = \frac{1}{2} \log_2^+(\text{Var}(Y | S)/D) = \frac{1}{2} \log_2^+(1/(2D))$. *Achievability.* For $D \leq \frac{1}{2}$ take $\hat{Y} = (1-D)X_1 + DX_2 + N$: distortion $D^2 + D^2 + D(1-2D) = D$; $\text{Var}(\hat{Y}) = 1-D$, $\text{Cov}(\hat{Y}, S) = 1$, $\text{Var}(\hat{Y} | S) = 1-D-\frac{1}{2} = \frac{1}{2}-D$, so $I(X; \hat{Y} | S) = \frac{1}{2} \log_2 \frac{(1-2D)/2}{D(1-2D)} = \frac{1}{2} \log_2 \frac{1}{2D}$. For $D \geq \frac{1}{2}$, $\hat{Y} = S/2$ is a deterministic function of X , has distortion $\mathbb{E}((X_1 - X_2)/2)^2 = \frac{1}{2} \leq D$, and is constant given S , so $I(X; \hat{Y} | S) = 0$.

(iii) Subtract. □

The closed form occupies the rest of this section. By Lemma 10 we restrict to T -channels, $T = (Y, V)$, with scalar reproduction $\hat{Y}$. We first fix the working coordinates. Every jointly Gaussian test channel takes the form $\hat{Y} = aY + bV + N$ with N independent noise of variance ν , and every admissible channel, Gaussian or not, is measured through the same three moments of its regression on T : the unconditional signal part $Q_0 = \text{Var}(aY + bV)$, the S -conditional signal part $Q_1 = \text{Var}(aY + bV | S)$, and the noise-free distortion contribution $h = \text{Var}((1-a)Y - bV)$, so that $\mathbb{E}(Y - \hat{Y})^2 = h + \nu$. Everything from here through Section VI is expressed in (a, b, ν) and these moments. The converse rests on one estimation-theoretic lemma. We state it for vector reproductions and vector side information, in the generality that Sections V and VI reuse.

Lemma 12. *Let (T, S) be jointly Gaussian and centered, $T \in \mathbb{R}^p$, $S \in \mathbb{R}^r$, with $\text{Cov}((T, S)) \succ 0$. Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be any centered random vector with finite second moments, and define*

$$A := \text{Cov}(\hat{\mathbf{Y}}, T) \Sigma_T^{-1} \in \mathbb{R}^{m \times p}, \quad N' := \hat{\mathbf{Y}} - AT, \quad \Sigma_N := \text{Cov}(N') = \text{Cov}(\hat{\mathbf{Y}}) - A \Sigma_T A^\top.$$

Assume $\Sigma_N \succ 0$ and $\text{Cov}(N', S) = 0$. Then

$$I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{\det(A \Sigma_{T|S} A^\top + \Sigma_N)}{\det \Sigma_N}, \quad (17)$$

where $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS} \Sigma_S^{-1} \Sigma_{ST}$ and Σ_e is the error covariance of the best linear estimator of T from $(\hat{\mathbf{Y}}, S)$. With S deleted throughout, the hypothesis $\text{Cov}(N', S) = 0$ becomes vacuous and the same argument gives

$$I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log_2 \frac{\det(A \Sigma_T A^\top + \Sigma_N)}{\det \Sigma_N}. \quad (18)$$

The jointly Gaussian channel $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$ independent of (T, S) , realizes the same second moments and attains both bounds with equality simultaneously. Conversely, (17) holds with equality only if the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is almost surely Gaussian with mean the best linear estimate and covariance Σ_e ; likewise for (18) with S deleted.

Proof. Step 1: the divergence bound. On standard Borel spaces, conditional mutual information is the expected divergence of the conditional law of T given $(\hat{\mathbf{Y}}, S)$ from the conditional law given S [7], [6]: $I(T; \hat{\mathbf{Y}} | S) =$

$\mathbb{E} D(P_{T|\hat{\mathbf{Y}}, S} \| P_{T|S})$, with the expectation over $(\hat{\mathbf{Y}}, S)$ and the value $+\infty$ allowed. Fix $(\hat{y}, s)$ outside a null set, write $P = P_{T|\hat{\mathbf{Y}}=\hat{y}, S=s}$, let μ and C be its mean and covariance, let $P_G = \mathcal{N}(\mu, C)$, and let $Q = P_{T|S=s} = \mathcal{N}(\mu_S(s), \Sigma_{T|S})$. If P has no Lebesgue density, $D(P\|Q) = \infty$. Otherwise, because $\log(dP_G/dQ)$ is a quadratic polynomial in t and P and P_G share their first and second moments, $\mathbb{E}_P \log(dP_G/dQ) = \mathbb{E}_{P_G} \log(dP_G/dQ)$, so $D(P\|Q) = D(P\|P_G) + D(P_G\|Q) \geq D(P_G\|Q)$, with equality if and only if $P = P_G$. In nats, $D(P_G\|Q) = \frac{1}{2} [\text{tr}(\Sigma_{T|S}^{-1}(C + \delta\delta^\top)) - p + \ln \det \Sigma_{T|S} - \ln \det C]$ with $\delta = \mu - \mu_S(s)$, and $-\ln \det C = +\infty$ if C is singular. Take expectations. By the law of total covariance, $\mathbb{E}[C + \delta\delta^\top | S] = \Sigma_{T|S}$, so the trace term averages to p . By concavity of $\ln \det$ and Jensen's inequality, $\mathbb{E}[-\ln \det C] \geq -\ln \det \mathbb{E}[C]$, and $\mathbb{E}[C] = \text{Cov}(T - \mathbb{E}[T | \hat{\mathbf{Y}}, S]) \preceq \Sigma_e$, because the conditional mean has error covariance no larger than that of any linear estimator. Hence $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log_2(\det \Sigma_{T|S} / \det \Sigma_e)$, the inequality in (17) once Step 2 evaluates $\det \Sigma_e$. Equality holds only if $P = P_G$ almost surely, $C = \Sigma_e$ almost surely (the equality case of Jensen's inequality, $\ln \det$ being strictly concave), and $\mathbb{E}[T | \hat{\mathbf{Y}}, S]$ is the best linear estimate almost surely, since $\text{Cov}(T - \hat{T}) = \mathbb{E}[C] + \text{Cov}(\mathbb{E}[T | \hat{\mathbf{Y}}, S] - \hat{T})$; the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is then $\mathcal{N}(\hat{T}, \Sigma_e)$, so $T - \hat{T}$ is Gaussian and independent of $(\hat{\mathbf{Y}}, S)$. The S -deleted case is identical with $Q = \mathcal{N}(0, \Sigma_T)$.

Step 2: the determinant identity. Write $\Omega := \text{Cov}((\hat{\mathbf{Y}}, S))$ and $J := \text{Cov}(T, (\hat{\mathbf{Y}}, S)) \in \mathbb{R}^{p \times (m+r)}$. First, $\Omega \succ 0$: by the Schur-complement criterion on

$$\Omega = \begin{bmatrix} \text{Cov}(\hat{\mathbf{Y}}) & \text{Cov}(\hat{\mathbf{Y}}, S) \\ \text{Cov}(S, \hat{\mathbf{Y}}) & \Sigma_S \end{bmatrix},$$

positivity is equivalent to $\Sigma_S \succ 0$ together with $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) \succ 0$; the first holds because $\text{Cov}(T, S) \succ 0$, and the second is evaluated below to equal $A\Sigma_{T|S}A^\top + \Sigma_N \succeq \Sigma_N \succ 0$. The best linear estimator of T from $(\hat{\mathbf{Y}}, S)$ is $\hat{T} = J\Omega^{-1}(\hat{\mathbf{Y}}, S)$, with error covariance $\Sigma_e = \Sigma_T - J\Omega^{-1}J^\top$. Let $K := \text{Cov}((T, S, \hat{\mathbf{Y}}))$. The residual N' is uncorrelated with T by construction of the regression and with S by hypothesis, and the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block lower triangular with unit diagonal, hence determinant preserving:

$$\det K = \det \text{Cov}((T, S, N')) = \det \begin{bmatrix} \Sigma_T & \Sigma_{TS} & 0 \\ \Sigma_{ST} & \Sigma_S & 0 \\ 0 & 0 & \Sigma_N \end{bmatrix} = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N,$$

using in the last step the Schur factorization $\det \text{Cov}(T, S) = \det \Sigma_S \det \Sigma_{T|S}$. The same Schur factorization applied to K in the other block order gives $\det K = \det \Omega \det \Sigma_e$, and applied to Ω itself gives

$$\det \Omega = \det \Sigma_S \cdot \det \left(\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) \right).$$

It remains to evaluate the conditional block. From $\hat{\mathbf{Y}} = AT + N'$ with $\text{Cov}(N', T) = 0$ and $\text{Cov}(N', S) = 0$, $\text{Cov}(\hat{\mathbf{Y}}) = A\Sigma_TA^\top + \Sigma_N$ and $\text{Cov}(\hat{\mathbf{Y}}, S) = A\Sigma_{TS}$, so

$$\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) = A(\Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST})A^\top + \Sigma_N = A\Sigma_{T|S}A^\top + \Sigma_N.$$

Combining the three displays,

$$\det \Sigma_e = \frac{\det K}{\det \Omega} = \frac{\det \Sigma_{T|S} \det \Sigma_N}{\det (A\Sigma_{T|S}A^\top + \Sigma_N)},$$

which is the identity in (17); in particular $\Sigma_e \succ 0$, so Step 1's bound is finite. Deleting S (formally $r = 0$: $K = \text{Cov}(T, \hat{\mathbf{Y}})$, $\Omega = \text{Cov}(\hat{\mathbf{Y}})$) gives $\det \Sigma_{e0} = \det \Sigma_T \det \Sigma_N / \det (A\Sigma_TA^\top + \Sigma_N)$ and hence (18).

Step 3: equality. For the stated Gaussian channel the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is $\mathcal{N}(\hat{T}, \Sigma_e)$ for every realization, so every inequality of Step 1 is an equality and (17) holds with equality; likewise for (18). The channel's moments match those of the given channel: $\text{Cov}(AT + N, T) = A\Sigma_T$ and $\text{Cov}(AT + N) = A\Sigma_TA^\top + \Sigma_N$. $\square$

Remark 13. If Σ_N is singular, some combination $\eta^\top \hat{\mathbf{Y}}$ equals $\eta^\top AT$ almost surely. If $\eta^\top A \neq 0$ the reproduction reveals a noiseless linear functional of the continuous T , so $I(T; \hat{\mathbf{Y}} | S) = \infty$ and every bound below holds trivially; if $\eta^\top A = 0$ that combination of the reproduction is zero and may be dropped, reducing m ; iterate until $\Sigma_N \succ 0$. All uses of Lemma 12 below may therefore assume $\Sigma_N \succ 0$.

Theorem 14. With $s = 1 + \tau^2$, the context-noise parameter of Section III, for $0 < D < 1$:

(a) The closed form.

$$L(D) = \frac{1}{2} \log_2 g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds}, \quad (19)$$

the larger root of the quadratic

$$P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2). \quad (20)$$

Since $P(1) = (D - 1)\tau^2 < 0$, exactly one root exceeds 1, so g^* is well defined. The optimum is attained by the jointly Gaussian channel

$$\hat{Y} = aY + bV + N, \quad a = \frac{g^* - 1}{g^*}, \quad b = \frac{(g^* - 1)\rho}{g^*k}, \quad k = g^*s - 1, \quad N \sim \mathcal{N}\left(0, \frac{Q_1}{g^* - 1}\right), \quad (21)$$

where $Q_1 = a^2 + b^2 + 2ab\rho - (a\rho + b)^2/s$ is evaluated at these (a, b) .

(b) The region. The union of the quadrants above the single-letter rate-content pairs $(I(T; \hat{Y}), I(T; \hat{Y} | S))$ over admissible test channels at distortion D , which Theorem 29 identifies with the operational region $\mathcal{RW}(D)$, is the closed convex set whose Pareto frontier is traced by $\alpha \in [0, 1]$ as follows. Let $(a, b, \nu, \gamma_0, \gamma_1)$, with $\nu > 0$ and $\gamma_0, \gamma_1 \in (0, 1)$, solve the stationarity system

$$a = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1, \quad \mu_c = \frac{a\rho}{s - (1 - \alpha)\gamma_1}, \quad b = (1 - \alpha)\gamma_1\mu_c, \quad (22)$$

$$\nu = D - h(a, b), \quad \gamma_0 = \frac{\nu}{Q_0 + \nu}, \quad \gamma_1 = \frac{\nu}{Q_1 + \nu}. \quad (23)$$

The solution exists and is unique at every α (Proposition 18 in Section V; at $\alpha = 0$ uniqueness also follows from part (a)), and the frontier point is

$$(R(\alpha), L(\alpha)) = \left(\frac{1}{2} \log_2 \frac{1}{\gamma_0}, \frac{1}{2} \log_2 \frac{1}{\gamma_1}\right),$$

achieved by $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, \nu)$. Endpoints: $\alpha = 1$ recovers the classical reverse channel ($b = 0$, $a = 1 - D$, $\gamma_0 = D$, $R = \frac{1}{2} \log_2(1/D)$); $\alpha = 0$ recovers the channel and value of part (a) ($\gamma_1 = 1/g^*$).

Part (a) is proved next. Part (b) is proved in Section V, after Proposition 18 there supplies the uniqueness it uses.

Proof of part (a). Moments. By Lemma 10 and the scalar-reproduction reduction of Section III, restrict to channels $p(\hat{y} | t)$. Fix an admissible channel with $\mathbb{E}(Y - \hat{Y})^2 \leq D$. Centering $\hat{Y}$ leaves $I(T; \hat{Y} | S)$ unchanged and can only decrease the distortion, so take $\mathbb{E}\hat{Y} = 0$. Define the regression coordinates

$$\begin{pmatrix} a \\ b \end{pmatrix} := \Sigma_T^{-1} \begin{pmatrix} \mathbb{E}[Y\hat{Y}] \\ \mathbb{E}[V\hat{Y}] \end{pmatrix}, \quad Q_0 := a^2 + b^2 + 2ab\rho = \text{Var}(aY + bV), \quad \nu := \text{Var} \hat{Y} - Q_0,$$

so that $\mathbb{E}[Y\hat{Y}] = a + b\rho$ and $\mathbb{E}[V\hat{Y}] = a\rho + b$; $\nu \geq 0$ is the residual variance of $\hat{Y}$ after linear regression on T . If $\nu = 0$, Remark 13 applies: either $(a, b) \neq (0, 0)$ and $I(T; \hat{Y} | S) = \infty$, or $(a, b) = (0, 0)$ and $\hat{Y} = 0$ with distortion $1 > D$. So take $\nu > 0$.

The Markov moment identity. The residual $N' = \hat{Y} - aY - bV$ is uncorrelated with T by construction. It is also uncorrelated with $S = V + U$: it is uncorrelated with V , a coordinate of T , and $\mathbb{E}[UN'] = 0$ because U is independent of $(X, \hat{Y})$ by the mutual independence of X , U , and the channel randomness. Hence $\text{Cov}(N', S) = 0$, which is the hypothesis of Lemma 12, and which restates the moment identity $\mathbb{E}[S\hat{Y}] = \mathbb{E}[V\hat{Y}]$.

The converse bound, written out. Here $p = 2$, $m = 1$, $r = 1$, $A = (a, b)$, $\Sigma_N = \nu$, and

$$\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \Sigma_{TS} = \begin{bmatrix} \rho \\ 1 \end{bmatrix}, \quad \Sigma_S = s,$$

$$\Sigma_{T|S} = \Sigma_T - \frac{1}{s} \Sigma_{TS} \Sigma_{TS}^T = \begin{bmatrix} 1 - \rho^2/s & \rho\tau^2/s \\ \rho\tau^2/s & \tau^2/s \end{bmatrix}, \quad \det \Sigma_{T|S} = \frac{\tau^2(1 - \rho^2)}{s} > 0.$$

The quadratic form in (17) is

$$A\Sigma_{T|S}A^\top = Q_0 - \frac{(a\rho + b)^2}{s} = \text{Var}(aY + bV | S) =: Q_1,$$

and, explicitly, $\Omega = \text{Cov}((\hat{Y}, S)) = \begin{bmatrix} Q_0 + \nu & a\rho + b \\ a\rho + b & s \end{bmatrix}$ with $\det \Omega = s(Q_0 + \nu) - (a\rho + b)^2 = s(Q_1 + \nu)$, while $\det \text{Cov}((T, S, \hat{Y})) = s \det \Sigma_{T|S} \cdot \nu$, so $\det \Sigma_e = \nu \det \Sigma_{T|S} / (Q_1 + \nu)$ and Lemma 12 reads

$$I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu}. \quad (24)$$

The distortion is a function of the same moments:

$$\mathbb{E}(Y - \hat{Y})^2 = 1 - 2(a + b\rho) + Q_0 + \nu = h(a, b) + \nu, \quad h(a, b) := (1 - a)^2 - 2(1 - a)b\rho + b^2, \quad (25)$$

and $h(a, b) = \text{Var}((1 - a)Y - bV) \geq 0$. Therefore

$$L(D) \geq \inf \left\{ B(a, b, \nu) := \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu} : h(a, b) + \nu \leq D, \nu > 0 \right\}, \quad (26)$$

and it suffices to evaluate the right side; achievability below matches it.

Activity, coercivity, existence. At fixed (a, b) with $Q_1 > 0$, B is strictly decreasing in ν ($\partial_\nu [\ln(Q_1 + \nu) - \ln \nu] = (Q_1 + \nu)^{-1} - \nu^{-1} < 0$), so the infimum is unchanged by imposing the active constraint $\nu = D - h(a, b)$. On the feasible set, (a, b) is confined to the compact set $\{h \leq D\}$; since $\Sigma_{T|S} \succ 0$, the form $Q_1 = (a, b) \Sigma_{T|S} (a, b)^\top$ vanishes only at $(a, b) = (0, 0)$, where $h = 1 > D$, so by continuity $Q_1 \geq q_0 > 0$ on the feasible set. Hence $B \geq \frac{1}{2} \log_2 ((q_0 + \nu)/\nu) \rightarrow \infty$ as $\nu \rightarrow 0$, the sublevel sets of the reduced objective

$$\Phi(a, b) := \frac{1}{2} \log_2 \frac{Q_1(a, b) + \nu(a, b)}{\nu(a, b)}, \quad \nu(a, b) = D - h(a, b),$$

are compact, and the infimum in (26) is attained at some (a, b) with $h(a, b) < D$: an interior stationary point of Φ .

The stationarity system is linear in (a, b) at fixed g . At a stationary point write $g := (Q_1 + \nu)/\nu$ and $\mu_c := (a\rho + b)/s$; note $g > 1$ because $Q_1 > 0$. The partial derivatives of the three quadratics are

$$\begin{aligned} \partial_a Q_0 &= 2(a + b\rho), & \partial_b Q_0 &= 2(b + a\rho), & \partial_a h &= \partial_a Q_0 - 2, & \partial_b h &= \partial_b Q_0 - 2\rho, \\ \partial_a Q_1 &= \partial_a Q_0 - 2\rho\mu_c, & \partial_b Q_1 &= \partial_b Q_0 - 2\mu_c, \end{aligned}$$

hence the four gradient identities

$$\partial_a(Q_0 - h) = 2, \quad \partial_b(Q_0 - h) = 2\rho, \quad \partial_a(Q_1 - h) = 2(1 - \rho\mu_c), \quad \partial_b(Q_1 - h) = 2(\rho - \mu_c). \quad (27)$$

(Only the last two enter here; all four enter the weighted system of Section V.) With $\nu = D - h$, so $\partial_x \nu = -\partial_x h$, stationarity of $2 \ln 2 \cdot \Phi = \ln(Q_1 + \nu) - \ln \nu$ in $x \in \{a, b\}$ reads

$$\frac{\partial_x(Q_1 - h)}{Q_1 + \nu} + \frac{\partial_x h}{\nu} = 0, \quad \text{that is,} \quad \frac{1}{g} \partial_x(Q_1 - h) + \partial_x h = 0.$$

By (27) and $\partial_a h = 2(a + b\rho - 1)$, $\partial_b h = 2(a\rho + b - \rho)$, the two equations are

$$a + b\rho = 1 - \frac{1 - \rho\mu_c}{g}, \quad a\rho + b = \rho - \frac{\rho - \mu_c}{g}. \quad (28)$$

Substituting $\mu_c = (a\rho + b)/s$ makes (28) a 2×2 linear system in (a, b) whose coefficients depend on g alone, and it solves in closed form. The second equation reads $gs\mu_c - \mu_c = \rho(g - 1)$, so with $k := gs - 1 > 0$,

$$\mu_c = \frac{\rho(g - 1)}{k}. \quad (29)$$

Subtracting ρ times the first equation of (28) from the second gives $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$ on the left and $\mu_c(1 - \rho^2)/g$ on the right, so $b = \mu_c/g$; the first equation then yields $a = 1 - (1 - \rho\mu_c)/g - b\rho = 1 - 1/g$. The stationary channel coefficients at level g are therefore

$$1 - a = \frac{1}{g}, \quad b = \frac{\mu_c}{g} = \frac{(g - 1)\rho}{gk}. \quad (30)$$

Reduction to the quadratic. At the values (30), $(1 - a)Y - bV = (Y - \mu_c V)/g$, so

$$h = \frac{1 - 2\rho\mu_c + \mu_c^2}{g^2}, \quad (31)$$

and, writing $aY + bV = Y - (Y - \mu_c V)/g$ and using the definition $a\rho + b = s\mu_c$,

$$Q_0 = 1 - \frac{2(1 - \rho\mu_c)}{g} + h, \quad Q_1 = Q_0 - \frac{(a\rho + b)^2}{s} = 1 - \frac{2(1 - \rho\mu_c)}{g} + h - s\mu_c^2.$$

The definition of g and the active constraint combine to $(g - 1)(D - h) = Q_1$. Substituting the display for Q_1 and collecting the h terms $(h + (g - 1)h = gh = (1 - 2\rho\mu_c + \mu_c^2)/g$ by (31)), the terms in $\rho\mu_c$ cancel and

$$1 + \frac{\mu_c^2 - 1}{g} - s\mu_c^2 = (g - 1)D. \quad (32)$$

Finally substitute (29). Multiplying (32) by gk^2 and using $k^2\mu_c^2 = \rho^2(g - 1)^2$,

$$gk^2 + \rho^2(g - 1)^2 - k^2 - gs\rho^2(g - 1)^2 = g(g - 1)Dk^2.$$

Since $1 - gs = -k$, the second and fourth terms on the left combine to $-\rho^2(g - 1)^2k$, so the left side is $(g - 1)k[k - \rho^2(g - 1)]$, and dividing by $(g - 1)k > 0$,

$$k - \rho^2(g - 1) = gDk \iff Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2) = 0,$$

that is, $P(g) = 0$. Every interior stationary point of Φ therefore sits at a root of P exceeding 1.

Root bracketing and uniqueness. $P(1) = Ds - (D + s - \rho^2) + (1 - \rho^2) = (D - 1)(s - 1) = (D - 1)\tau^2 < 0$ for $D < 1$, $\tau^2 > 0$. The leading coefficient Ds is positive and $P(0) = 1 - \rho^2 > 0$, so P has two real roots, exactly one exceeding 1 (the other lies in $(0, 1)$). Hence every stationary point has $g = g^*$, and by (30) the stationary (a, b) , and with it $\nu = Q_1/(g^* - 1)$, is unique. The attained minimum of (26) is $\frac{1}{2}\log_2 g^*$.

Achievability. The channel (21) has exactly the moments (30) with residual variance $\nu = Q_1/(g^* - 1)$. Its distortion is $h + \nu$, and $(g^* - 1)\nu = Q_1$ together with $P(g^*) = 0$ is, by the chain of equivalences above run backward, the statement $h + \nu = D$. The channel is jointly Gaussian with (T, S) , so the equality clause of Lemma 12 gives

$$I(T; \hat{Y} | S) = \frac{1}{2}\log_2 \frac{Q_1 + \nu}{\nu} = \frac{1}{2}\log_2 g^*.$$

Hence $L(D) = \frac{1}{2}\log_2 g^*$. □

Corollary 15. (i) At $\rho = 0$, $P(g) = (Dg - 1)(sg - 1)$ and $g^* = 1/D$: $L(D) = \frac{1}{2}\log_2(1/D)$ is the classical rate-distortion function, attained with $b = 0$ by the classical reverse test channel. (ii) As $\tau^2 \rightarrow 0$, $L(D) \rightarrow \frac{1}{2}\log_2^+((1 - \rho^2)/D)$, Gray's conditional rate-distortion function of the pair (Y, V) [4], matching Proposition 11; for $\tau^2 > 0$, $\rho \neq 0$, and $D < 1 - \rho^2$ the value lies strictly above Gray's. (iii) As $\rho^2 \rightarrow 1$, $L(D) \rightarrow \frac{1}{2}\log_2[(D + \tau^2)/(D(1 + \tau^2))]$, Steinberg's scalar Gaussian formula at the induced correlation $\rho_{YS}^2 = 1/s$ [12].

Proof. (i) $(Dg - 1)(sg - 1) = Dsg^2 - (D + s)g + 1 = P(g)|_{\rho=0}$. The roots are $1/D > 1$ and $1/s < 1$, so $g^* = 1/D$. At this root (21) gives $b = 0$, $a = 1 - D$, $Q_1 = (1 - D)^2$, and $\text{Var } N = (1 - D)^2/(1/D - 1) = D(1 - D)$: the classical reverse channel.

(ii) At $s = 1$, $P(g) = Dg^2 - (D + 1 - \rho^2)g + (1 - \rho^2) = (g - 1)(Dg - (1 - \rho^2))$, with discriminant $(D - (1 - \rho^2))^2$, so the larger root at $s = 1$ is exactly $\max\{1, (1 - \rho^2)/D\}$; the roots of P are continuous in s , and the limit follows. For $D < 1 - \rho^2$ the value is $\frac{1}{2}\log_2(\text{Var}(Y | V)/D)$ with $\text{Var}(Y | V) = 1 - \rho^2$, Gray's conditional function; for $D \geq 1 - \rho^2$ it is 0. Strictness for $\tau^2 > 0$: at $g_G = (1 - \rho^2)/D > 1$, $P(g_G) = -\rho^2\tau^2(1 - \rho^2)/D < 0$ when $\rho \neq 0$, so g_G lies between the two roots and $g^* > g_G$.

(iii) At $\rho^2 = 1$ the constant term of P vanishes: $P(g) = g(Dsg - (D + \tau^2))$, with roots 0 and $(D + \tau^2)/(Ds)$; the latter exceeds 1 because $(D + \tau^2) - Ds = \tau^2(1 - D) > 0$. By root continuity $g^* \rightarrow (D + \tau^2)/(Ds)$ as $\rho^2 \rightarrow 1$. Steinberg's scalar function at unit source variance and side-information correlation ρ_{YS}^2 is $\frac{1}{2}\log_2[(1 - \rho_{YS}^2 + \rho_{YS}^2 D)/D]$ [12]; at $\rho_{YS}^2 = \rho^2/s \rightarrow 1/s$ this is $\frac{1}{2}\log_2[(\tau^2 + D)/(sD)]$, the same value. □

Corollary 16. *Under the standing model, for every $\rho \neq 0$ and $0 < D < 1$, $L(D) < \frac{1}{2} \log_2 g_m$ with $g_m = (s - \rho^2 + \rho^2 D)/(sD)$, the value of Steinberg's scalar formula [12] on the reduced pair (Y, S) ; so a channel that depends on X through Y alone is strictly suboptimal.*

Proof. The marginalized class attains exactly $\frac{1}{2} \log_2 g_m$, the scalar-corner value of the reduction preceding Proposition 11 at $\rho_{Y|S}^2 = \rho^2/s$. Substituting into (20) gives $P(g_m) = \rho^2(1 - D)(1 - \rho^2)/s > 0$ and $g_m - 1 = (1 - D)(s - \rho^2)/(Ds) > 0$. Since $P(1) = (D - 1)\tau^2 < 0$, the two roots of P bracket 1; a point above 1 at which $P > 0$ lies above the larger root, so $g_m > g^*$. The same value $\frac{1}{2} \log_2 g_m$ reappears in Corollary 19 as the conditional content of the rate-optimal channel, which lies in the marginalized class. $\square$

V. THE RATE-CONTENT REGION AND MISALIGNMENT

Attaining the content endpoint of Theorem 14 costs rate. This section characterizes the full tradeoff between the two coordinates of (11) for the Gaussian pair source. Throughout, $|\rho| < 1$, $\tau^2 > 0$, $0 < D < 1$, and h, Q_0, Q_1, μ_c are as in the proof of Theorem 14. In the frontier system below, $\alpha \in [0, 1]$ weights rate against conditional content, and $\gamma_0 = \nu/(Q_0 + \nu)$ and $\gamma_1 = \nu/(Q_1 + \nu)$ are the two noise fractions the stationarity system balances, the shares of the reproduction's unconditional and S -conditional variance that its noise occupies: the frontier point at weight α is $(\frac{1}{2} \log_2(1/\gamma_0), \frac{1}{2} \log_2(1/\gamma_1))$, and the system couples the two fractions through one reproduction and one distortion constraint. The proof of part (b) uses the uniqueness of the weighted minimizer, so the whitened frame, Lemma 17, and Proposition 18 come first.

Uniqueness at every weight is a convexity statement in moment coordinates, stated in the whitened frame that Section VI also uses; we set it up once. Since $\Sigma_T \succ 0$ and $0 \prec \Sigma_{T|S} \preceq \Sigma_T$, there is a nonsingular W with

$$W\Sigma_TW^\top = I, \quad W\Sigma_{T|S}W^\top = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad 0 \prec \Lambda \preceq I \quad (33)$$

(factor $\Sigma_T = CC^\top$, diagonalize $C^{-1}\Sigma_{T|S}C^{-\top} = O\Lambda O^\top$, and set $W = O^\top C^{-1}$). Write $Z = WT$, so $\text{Cov } Z = I$; the described variable becomes $Y = y_0^\top Z$ with $y_0 = W^{-\top}e_Y$ and $|y_0|^2 = e_Y^\top \Sigma_T e_Y = 1$. A channel with regression vector $(a, b)^\top$ on T has regression vector $c = W^{-\top}(a, b)^\top$ on Z , and

$$Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad h = |y_0 - c|^2, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + \nu.$$

Here $p = 2$; Section VI uses $p = 1 + r$.

Lemma 17. *Let $\mathcal{Z} \subseteq \mathbb{R}^q$ be convex, $u : \mathcal{Z} \rightarrow \mathbb{R}^p$ affine, and $\sigma : \mathcal{Z} \rightarrow (0, \infty)$ concave. Then*

$$F(z) := \frac{u(z)u(z)^\top}{\sigma(z)}$$

is matrix-convex: $F(z_t) \preceq (1 - t)F(z_0) + tF(z_1)$ for all $z_0, z_1 \in \mathcal{Z}$, $t \in [0, 1]$, $z_t = (1 - t)z_0 + tz_1$.

Proof. For $\sigma > 0$ the Schur-complement criterion [38, Sec. A.5.5] states

$$\begin{bmatrix} G & u \\ u^\top & \sigma \end{bmatrix} \succeq 0 \iff G \succeq \frac{uu^\top}{\sigma},$$

so $F(z)$ is the smallest matrix admitted by the criterion at z . Fix z_0, z_1, t and abbreviate $u_i = u(z_i)$, $\sigma_i = \sigma(z_i)$, $F_i = F(z_i)$. The two positive semidefinite block matrices

$$\begin{bmatrix} F_0 & u_0 \\ u_0^\top & \sigma_0 \end{bmatrix} \succeq 0, \quad \begin{bmatrix} F_1 & u_1 \\ u_1^\top & \sigma_1 \end{bmatrix} \succeq 0$$

average, with weights $(1 - t, t)$, to

$$\begin{bmatrix} (1 - t)F_0 + tF_1 & u(z_t) \\ u(z_t)^\top & (1 - t)\sigma_0 + t\sigma_1 \end{bmatrix} \succeq 0,$$

using affinity of u in the off-diagonal blocks. Concavity gives $\sigma(z_t) \geq (1 - t)\sigma_0 + t\sigma_1$, and adding the positive semidefinite matrix $\text{diag}(0, \sigma(z_t) - (1 - t)\sigma_0 - t\sigma_1)$ raises the corner:

$$\begin{bmatrix} (1 - t)F_0 + tF_1 & u(z_t) \\ u(z_t)^\top & \sigma(z_t) \end{bmatrix} \succeq 0.$$

By the criterion again, $(1-t)F_0 + tF_1 \succeq u(z_t)u(z_t)^\top/\sigma(z_t) = F(z_t)$. $\square$

Proposition 18. *In the whitened frame (33), with $0 \prec \Lambda \preceq I$, $|y_0| = 1$, $D \in (0, 1)$, and any $\alpha \in [0, 1]$, the weighted program (36) (with $Q_0 = |c|^2$, $Q_1 = c^\top \Lambda c$, $h = |y_0 - c|^2$, minimized over (c, ν)) has a unique minimizer, and every stationary point with $\nu > 0$ is that minimizer. Since the change of variables $c = W^{-\top}(a, b)^\top$ is a linear bijection, the same holds for the program in the coordinates of Theorem 14.*

Proof. Work in the moment coordinates $z = (c, v)$, $v = \text{Var } \hat{Y} = |c|^2 + \nu$, on the domain

$$\mathcal{D} = \{(c, v) : v > |c|^2, 1 - 2y_0^\top c + v \leq D\},$$

which is convex: $v > |c|^2$ is the strict epigraph of a convex function, and the distortion constraint $h + \nu = 1 - 2y_0^\top c + v \leq D$ is linear in z . On $\mathcal{D}$ define $\sigma := v - c^\top(I - \Lambda)c = Q_1 + \nu > 0$ and

$$\begin{aligned} B_0 &= \frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu} = -\frac{1}{2} \log_2 \det \left(I - \frac{cc^\top}{\nu} \right), \\ B_1 &= \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu} = \frac{1}{2} \log_2 \det \Lambda - \frac{1}{2} \log_2 \det \left(\Lambda - \frac{(\Lambda c)(\Lambda c)^\top}{\sigma} \right); \end{aligned}$$

the first equality holds because $\det(I - cc^\top/\nu) = 1 - |c|^2/\nu = \nu/\nu$, the second because $\det(\Lambda - (\Lambda c)(\Lambda c)^\top/\sigma) = \det \Lambda (1 - c^\top \Lambda c/\sigma) = \det \Lambda \cdot \nu/\sigma$.

Convexity. Apply Lemma 17 twice. For B_0 : $z \mapsto c$ is affine and $z \mapsto v$ is affine, hence concave and positive on $\mathcal{D}$, so $G_0(z) = cc^\top/\nu$ is matrix-convex. Moreover $I - G_0 \succ 0$ on $\mathcal{D}$, hence invertible: by the Schur criterion, $I \succ cc^\top/\nu$ is equivalent to $v - |c|^2 > 0$, and $v - |c|^2 = \nu > 0$ on $\mathcal{D}$ by construction. For B_1 : $z \mapsto \Lambda c$ is affine and $z \mapsto \sigma = v - c^\top(I - \Lambda)c = Q_1 + \nu$ is concave (affine minus a convex quadratic, $I - \Lambda \succeq 0$) and positive on $\mathcal{D}$ ($Q_1 \geq 0$, $\nu > 0$), so $G_1(z) = (\Lambda c)(\Lambda c)^\top/\sigma$ is matrix-convex, $\Lambda - G_1$ is matrix-concave, and $\Lambda - G_1 \succ 0$ because $c^\top \Lambda c = Q_1 < Q_1 + \nu = \sigma$. The map $\Sigma \mapsto -\log_2 \det \Sigma$ on positive definite matrices is convex and decreasing in the Löwner order; for the latter, if $0 \prec A \preceq B$ then $B^{-1/2}AB^{-1/2} \preceq I$ has eigenvalues in $(0, 1]$, so $\det A/\det B = \det(B^{-1/2}AB^{-1/2}) \leq 1$, that is, $\det A \leq \det B$. Composing a convex antitone function with a matrix-concave map preserves convexity: for matrix-concave $F \succ 0$,

$$-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq (1-t)(-\log \det F(z_0)) + t(-\log \det F(z_1)).$$

Hence B_0 , B_1 , and every $B_\alpha := \alpha B_0 + (1-\alpha)B_1$ are convex on $\mathcal{D}$.

Reduction to the active slice. On $\mathcal{D}$, feasibility forces $c \neq 0$: from $v > |c|^2 \geq 0$ and $1 - 2y_0^\top c + v \leq D$,

$$2y_0^\top c \geq 1 - D + v > 1 - D > 0. \quad (34)$$

At fixed $c \neq 0$ both B_0 and B_1 strictly decrease in v ($\partial_v[\ln v - \ln \nu] = v^{-1} - \nu^{-1} < 0$ and $\partial_v[\ln \sigma - \ln \nu] = \sigma^{-1} - \nu^{-1} < 0$, since $Q_0 \geq Q_1 > 0$), so every minimizer of B_α lies on the active slice $v = D - 1 + 2y_0^\top c$, an affine section of $\mathcal{D}$. On the slice B_α is convex, so its stationary points are exactly its global minimizers and the minimizer set is convex; coercivity gives attainment, since on the slice c ranges over the closed ball $|y_0 - c|^2 \leq D$, which excludes $c = 0$ because $D < 1$, so $Q_1 \geq \lambda_{\min}|c|^2$ is bounded away from zero and $B_\alpha \rightarrow \infty$ as $\nu \rightarrow 0$.

Strictness. Suppose $z_0 \neq z_1$ are both minimizers on the slice, and let z_t run over the segment, along which v and c are affine in t . Then B_α is affine on the segment, and since B_0 , B_1 are convex and their weighted sum is affine, each B_i carrying nonzero weight is affine on the segment. We use twice the following fact: for an affine positive definite matrix segment $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$, the function $t \mapsto -\log \det \Sigma_t$ has second derivative $\text{tr}[(\Sigma_t^{-1} \dot{\Sigma})^2] \geq 0$, which vanishes only if $\dot{\Sigma} = 0$, because $\Sigma_t^{-1/2} \dot{\Sigma} \Sigma_t^{-1/2}$ is symmetric and a symmetric matrix whose square has zero trace is zero. Moreover, if a matrix-concave F has $-\log \det F$ affine on the segment, then squeezing $-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq$ the affine interpolant forces both inequalities to be equalities; the second forces $F(z_0) = F(z_1) =: \bar{F}$ by the fact just stated, and the first then forces $F(z_t) \succeq \bar{F}$ with $\det F(z_t) = \det \bar{F}$, hence $F(z_t) = \bar{F}$ for all t (if $A \succeq B \succ 0$ and $\det A = \det B$ then $A = B$).

Case $\alpha > 0$, where strictness comes from the forced constancy of the rank-one ratio G_0 along the segment: B_0 is affine, so $G_0(z_t) = c(t)c(t)^\top/\nu(t)$ is constant, a rank-one matrix $\bar{G}_0 \neq 0$ by (34). Then $c(t) = \theta(t)e$ for a fixed unit vector e , with θ affine (a coordinate of the affine c), and $\theta(t)^2 = \nu(t)(e^\top \bar{G}_0 e)$ with ν affine. An affine function whose square is affine is constant, so θ , hence c , hence $v = D - 1 + 2y_0^\top c$, is constant: $z_0 = z_1$, a contradiction.

Case $\alpha < 1$, where strictness comes from the constancy of the rank-one ratio G_1 played against the concavity of σ : B_1 is affine, so $G_1(z_t) = (\Lambda c)(\Lambda c)^\top / \sigma$ is constant and nonzero. Then $\Lambda c(t) = \theta(t)e$ with θ affine, and $\theta(t)^2 = \sigma(t)(e^\top \bar{G}_1 e)$. Along the segment $\sigma(t) = v(t) - c(t)^\top (I - \Lambda)c(t)$ is a concave quadratic in t , while θ^2 is a convex quadratic; equality up to the positive constant $e^\top \bar{G}_1 e$ forces both to be affine, so θ is an affine function with affine square, hence constant. Then Λc is constant, c is constant (Λ nonsingular), and $z_0 = z_1$ as before, a contradiction.

Every $\alpha \in [0, 1]$ falls in at least one case, so the minimizer is unique.

We record one further fact for the clean-context boundary. At $\alpha = 1$ the argument is Λ -free: the objective is B_0 alone, whose convexity (Lemma 17 with $u(z) = c$ and $\sigma = v$), strict decrease in v at $c \neq 0$, coercivity through $Q_0 \geq q_0 > 0$, and strictness case use only c , v , and the domain $\mathcal{D}$, never Λ . The uniqueness of the rate minimizer, equivalently of the moment triple $(a, b, \nu) = (1 - D, 0, D(1 - D))$, therefore holds for arbitrary side information, including the boundary $\tau^2 = 0$, where $\Sigma_{T|S}$ is singular and the whitening (33) of the conditional covariance is unavailable. $\square$

We now prove part (b) of Theorem 14, the region and its frontier.

Proof of part (b). Convexity and scalarization. Both coordinates are convex in the test channel, by Lemma 3 in the general-alphabet form of Remark 4. The achievable set of pairs at distortion D is therefore convex and upward closed in each coordinate (and closed, by the closedness step of Appendix A), so every point of its Pareto frontier minimizes one of the weighted objectives

$$J_\alpha := \alpha I(T; \hat{Y}) + (1 - \alpha) I(T; \hat{Y} | S), \quad \alpha \in [0, 1],$$

over admissible channels, by the supporting-hyperplane argument. At the interior weights each J_α -minimizer is a frontier point; at the endpoint weights $\alpha \in \{0, 1\}$ this a priori fixes only one coordinate of the minimizer, and the value of the other coordinate is pinned by the uniqueness of the minimizing moments (Proposition 18; the argument is run in detail in the proof of Corollary 19).

The two determinant identities. Fix an admissible channel, centered without loss, with regression coordinates (a, b, ν) , $\nu > 0$, as in the proof of part (a); the Markov moment identity $\text{Cov}(N', S) = 0$ holds as there. Lemma 12 applies twice to the same three moments, once with the side information S and once with S deleted:

$$I(T; \hat{Y}) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_T}{\det \Sigma_{e0}} = \frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu}, \quad I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu}, \quad (35)$$

where Σ_{e0} is the error covariance of the best linear estimate of T from $\hat{Y}$ alone. The conditional identity is written out in the proof of part (a). The rate identity, in full: with $\hat{c} := \mathbb{E}[T\hat{Y}] = \Sigma_T(a, b)^\top$ and $\text{Var } \hat{Y} = Q_0 + \nu$,

$$\Sigma_{e0} = \Sigma_T - \frac{\hat{c}\hat{c}^\top}{Q_0 + \nu}, \quad \det \Sigma_{e0} = \det \Sigma_T \left(1 - \frac{\hat{c}^\top \Sigma_T^{-1} \hat{c}}{Q_0 + \nu}\right) = \det \Sigma_T \frac{\nu}{Q_0 + \nu},$$

by the rank-one determinant identity $\det(\Sigma - \hat{c}\hat{c}^\top/\nu) = \det \Sigma (1 - \hat{c}^\top \Sigma^{-1} \hat{c}/\nu)$ and $\hat{c}^\top \Sigma_T^{-1} \hat{c} = (a, b) \Sigma_T (a, b)^\top = Q_0$.

The weighted moment program. Combining, J_α is bounded below by

$$\min_{(a,b,\nu)} \varphi_\alpha(a, b, \nu) := \frac{\alpha}{2} \log_2 \frac{Q_0 + \nu}{\nu} + \frac{1 - \alpha}{2} \log_2 \frac{Q_1 + \nu}{\nu} \quad \text{s.t.} \quad h(a, b) + \nu \leq D, \quad \nu > 0. \quad (36)$$

As in the proof of part (a): φ_α is strictly decreasing in ν at fixed $(a, b) \neq (0, 0)$, so the distortion constraint is active at the optimum; on the feasible set $Q_0 \geq Q_1 \geq q_0 > 0$, so $\varphi_\alpha \rightarrow \infty$ as $\nu \rightarrow 0$ and the minimum is attained at an interior stationary point of $\varphi_\alpha(a, b, D - h(a, b))$.

The gradient identities reduce the first-order system linearly. At a stationary point set $\gamma_0 = \nu/(Q_0 + \nu)$ and $\gamma_1 = \nu/(Q_1 + \nu)$, both in $(0, 1)$. For $x \in \{a, b\}$, using $\partial_x \nu = -\partial_x h$,

$$\alpha \frac{\partial_x(Q_0 - h)}{Q_0 + \nu} + (1 - \alpha) \frac{\partial_x(Q_1 - h)}{Q_1 + \nu} + \frac{\partial_x h}{\nu} = 0;$$

multiplying by $\nu/2$ and inserting the four gradient identities (27):

$$x = a : \quad \alpha \gamma_0 + (1 - \alpha) \gamma_1 (1 - \rho \mu_c) + (a + b\rho - 1) = 0, \quad (37)$$

$$x = b : \quad \alpha \gamma_0 \rho + (1 - \alpha) \gamma_1 (\rho - \mu_c) + (a\rho + b - \rho) = 0. \quad (38)$$

Subtract ρ times (37) from (38): the γ_0 terms cancel, the γ_1 terms give $(1 - \alpha)\gamma_1[(\rho - \mu_c) - \rho(1 - \rho\mu_c)] = -(1 - \alpha)\gamma_1\mu_c(1 - \rho^2)$, and the moment terms give $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$, so

$$b(1 - \rho^2) = (1 - \alpha)\gamma_1\mu_c(1 - \rho^2) \implies b = (1 - \alpha)\gamma_1\mu_c,$$

dividing by $1 - \rho^2 > 0$, which is in force since $|\rho| < 1$ throughout this section. Equation (37) then reads $a + b\rho = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1 + (1 - \alpha)\gamma_1\rho\mu_c = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1 + b\rho$, so $a = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1$. Finally $s\mu_c = a\rho + b = a\rho + (1 - \alpha)\gamma_1\mu_c$ rearranges to $\mu_c = a\rho/(s - (1 - \alpha)\gamma_1)$, the denominator being positive since $s > 1 > (1 - \alpha)\gamma_1$. This is (22); (23) restates the activity of the constraint and the definitions of (γ_0, γ_1) . Conversely, every solution of the system with $\nu > 0$ is a stationary point of the reduced program, and by Proposition 18 the program is convex in moment coordinates with a unique minimizer, so the system's solution is unique and is that minimizer.

Achievability. The jointly Gaussian channel $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, \nu)$, with the optimal (a, b, ν) realizes the same moments and, by the equality clause of Lemma 12, attains both bounds of (35) simultaneously: $I(T; \hat{Y}) = \frac{1}{2} \log_2(1/\gamma_0)$ and $I(T; \hat{Y} | S) = \frac{1}{2} \log_2(1/\gamma_1)$. Hence the value of (36) equals the minimum of J_α , the frontier point is as displayed, and the trace over $\alpha \in [0, 1]$ is the entire Pareto boundary.

Endpoints. At $\alpha = 1$, (22) gives $b = 0$ and $a = 1 - \gamma_0$; then $h = (1 - a)^2 = \gamma_0^2$, $Q_0 = a^2$, and $\gamma_0(Q_0 + \nu) = \nu$ forces $\gamma_0 a^2 = \nu(1 - \gamma_0) = \nu a$, so $\nu = \gamma_0 a$; combined with $\nu = D - \gamma_0^2$ this gives $\gamma_0(1 - \gamma_0) = D - \gamma_0^2$, that is, $\gamma_0 = D$, $a = 1 - D$, $\nu = D(1 - D)$: the classical reverse channel. At $\alpha = 0$ the γ_0 terms drop from (37)–(38), which become (28) with $g = 1/\gamma_1$; part (a) identifies $\gamma_1 = 1/g^*$ and the channel (21). $\square$

Corollary 19. *Within the standing model of Section III ($\tau^2 > 0$, $|\rho| < 1$, $0 < D < 1$), the frontier is nondegenerate if and only if $\rho^2 \in (0, 1)$. When $\rho^2 \in (0, 1)$: any channel attaining $L(D)$ has rate strictly above $R_{\min} = \frac{1}{2} \log_2(1/D)$, and any channel attaining $R_{\min}$ has conditional content strictly above $L(D)$. In particular, writing $L_R^*(D)$ for the conditional content of the rate-optimal channel and $R_L^*(D)$ for the rate of the content-optimal channel,*

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1 - D)(1 - \rho^2/s) + D}{D} > L(D),$$

the content of the classical reverse channel, which equals $\frac{1}{2} \log_2 g_m$, Steinberg's scalar formula on the reduced pair (Y, S) of Corollary 16: the rate-optimal channel ignores the context, and the content it pays is the marginalized-class optimum; and $R_L^(D) = \frac{1}{2} \log_2[(Q_0 + \nu)/\nu]$ evaluated at the channel of Theorem 14 strictly exceeds $R_{\min}$. When $\rho = 0$ the classical reverse channel attains both minima and the frontier is a single corner. The frontier also collapses in the limits $\rho^2 \rightarrow 1$ (where both endpoint excesses vanish; see the proof) and $\tau^2 \rightarrow \infty$ (where $L \rightarrow R$ coordinatewise); positivity of τ^2 , by contrast, is a modeling assumption and not necessary for the tradeoff (Proposition 20).*

Proof. By part (a) of Theorem 14, the moment program for the conditional coordinate alone has the unique minimizer (a^*, b^*, ν^*) of (30), and $b^* = (g^* - 1)\rho/(g^*k) \neq 0$ since $\rho \neq 0$, $g^* > 1$, $k > 0$. By the $\alpha = 1$ endpoint computation in the proof of part (b) (uniqueness by Proposition 18), the moment program for the rate coordinate alone has the unique minimizer $(1 - D, 0, D(1 - D))$, with $b = 0$. The two minimizers are distinct.

Let a channel attain $L(D)$. Writing B_0, B_1 for the two bounds of (35) as functions of the channel's moments, $L(D) = I(T; \hat{Y} | S) \geq B_1(\text{its moments}) \geq \min B_1 = L(D)$ forces equality throughout, so its moments minimize B_1 and equal (a^*, b^*, ν^*) . Its rate then satisfies $R \geq B_0(a^*, b^*, \nu^*) > \min B_0 = R_{\min}$: strict, because B_0 's minimizer is unique and differs from (a^*, b^*, ν^*) . Symmetrically, any rate-optimal channel carries the moments $(1 - D, 0, D(1 - D))$ and its content is at least B_1 there, which strictly exceeds $L(D)$; evaluating, $b = 0$ gives $Q_1 = (1 - D)^2(1 - \rho^2/s)$ and $\nu = D(1 - D)$, so $(Q_1 + \nu)/\nu = [(1 - D)(1 - \rho^2/s) + D]/D$, the displayed $L_R^*(D)$.

At $\rho = 0$, $b^* = 0$ and $g^* = 1/D$ (Corollary 15(i)) give $(a^*, \nu^*) = (1 - D, D(1 - D))$: the two minimizers coincide and the frontier is the corner $R_{\min} = L(D) = \frac{1}{2} \log_2(1/D)$. As $\rho^2 \rightarrow 1$ both endpoint excesses vanish, so the frontier collapses to a corner in the limit: by Corollary 15(iii), $L(D) \rightarrow \frac{1}{2} \log_2[(D + \tau^2)/(Ds)]$, and the displayed $L_R^*(D)$ has the same limit, since $(1 - D)(1 - \rho^2/s) + D \rightarrow (1 - D)\tau^2/s + D = (D + \tau^2)/s$; for the rate side, the channel (21) has, as $\rho \rightarrow 1$, $g^* \rightarrow (D + \tau^2)/(Ds)$ and

$$(a, b, \nu) \longrightarrow \left(\frac{\tau^2(1 - D)}{D + \tau^2}, \frac{D(1 - D)}{D + \tau^2}, D(1 - D) \right),$$

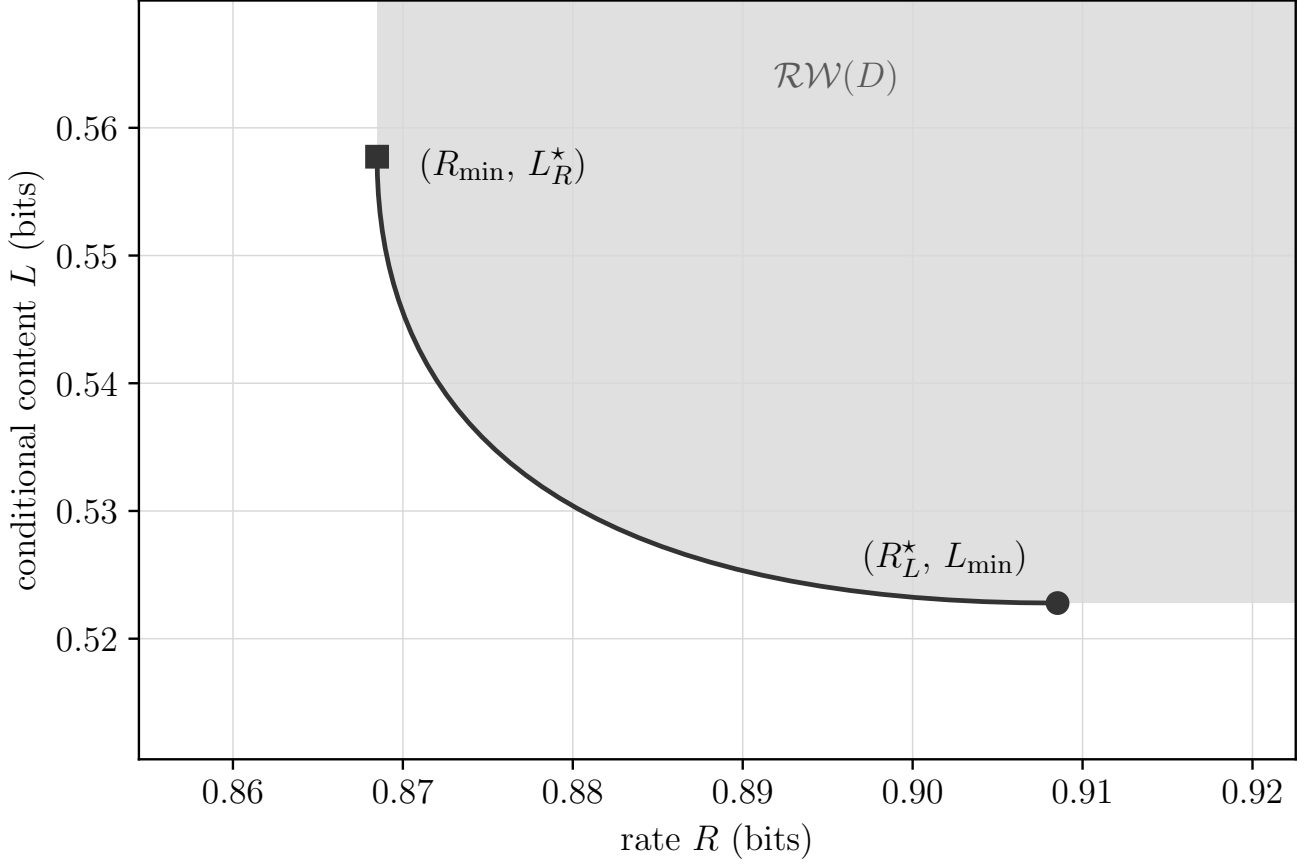


Fig. 2. The rate-content region $\mathcal{RW}(D)$ (shaded) and its Pareto frontier at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$, the second instance of Table II at $D = 0.3$, traced by the two-noise-fraction system of Theorem 14(b). The tradeoff is strict: the rate-optimal channel pays $L_R^*(D) - L(D) = 0.0349$ bits of excess conditional content, and the content-optimal channel pays $R_L^*(D) - R_{\min} = 0.0400$ bits of excess rate.

whence $a + b \rightarrow 1 - D$, $Q_0 \rightarrow (1 - D)^2$, and $R_L^*(D) = \frac{1}{2} \log_2[(Q_0 + \nu)/\nu] \rightarrow \frac{1}{2} \log_2(1/D) = R_{\min}$; for $\rho \rightarrow -1$ only the sign of b flips, leaving Q_0 and R_L^* unchanged. As $\tau^2 \rightarrow \infty$, $\Sigma_{T|S} \rightarrow \Sigma_T$ makes the two coordinates, and so the two endpoints, coincide in the limit. $\square$

Fig. 2 traces the frontier at the numerical instance of Corollary 19. The displayed excesses are representative of moderate parameters, not extremal. Over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$, the endpoint excesses reach at most $R_L^*(D) - R_{\min} = 0.1138$ bits, at $(\rho^2, \tau^2, D) = (0.5, 0.25, 0.5)$, and $L_R^*(D) - L(D) = 0.0770$ bits, at $(0.615, 0.25, 0.385)$. Toward the clean-context boundary the rate excess is unbounded: at $D = 1 - \rho^2$ the content-optimal channel pins the reproduction to the context as $\tau^2 \rightarrow 0$ and its rate diverges, reaching, for instance, $R_L^*(D) - R_{\min} = 1.537$ bits at $(0.5, 10^{-3}, 0.5)$; the content excess instead saturates there at $\frac{1}{2} \log_2(1 + \rho^2)$, the clean-boundary value of Proposition 20 evaluated at $D = 1 - \rho^2$. All quoted values are computed by the scripts of Appendix B. Fig. 3 repeats the construction at $\rho^2 \in \{0.2, 0.5, 0.8\}$ with $\tau^2 = 0.5$ and $D = 0.3$; the frontier is a nondegenerate curve at each, as Corollary 19 states.

Proposition 20. *The strict tradeoff extends to the boundary $\tau^2 = 0$ excluded by the standing model. Let $\tau^2 = 0$, so $S = V$, and let $\rho^2 \in (0, 1)$, $0 < D < 1$. The minimal conditional content is*

$$L_{\min}(D) = \frac{1}{2} \log_2^+ \frac{1 - \rho^2}{D}, \quad (39)$$

while every channel attaining the minimal rate $R_{\min} = \frac{1}{2} \log_2(1/D)$, in particular the classical reverse channel

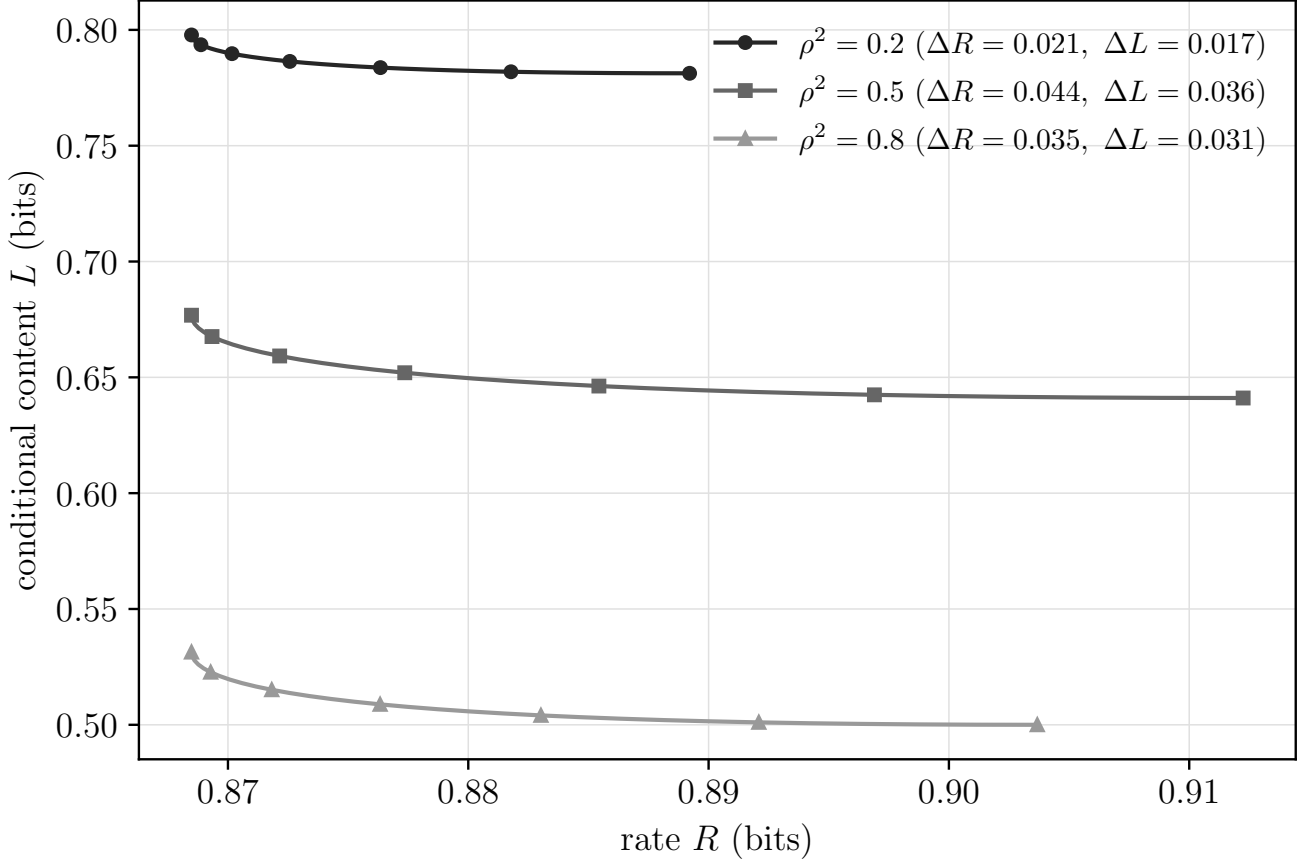


Fig. 3. The frontier at three correlations, fixed $(\tau^2, D) = (0.5, 0.3)$. The endpoint gap between the rate-optimal and content-optimal channels widens from $\rho^2 = 0.2$ ($\Delta R = 0.021$, $\Delta L = 0.017$ bits) to $\rho^2 = 0.5$ ($\Delta R = 0.044$, $\Delta L = 0.036$) and eases at $\rho^2 = 0.8$ ($\Delta R = 0.035$, $\Delta L = 0.031$); the tradeoff vanishes at $\rho^2 = 0$, is widest at intermediate correlation, and closes again toward the $\rho^2 \rightarrow 1$ boundary. Marker shape distinguishes the curves.

$\hat{Y}_R = (1 - D)Y + N$ with $N \sim \mathcal{N}(0, D(1 - D))$ independent of X , has conditional content exactly

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1 - D)(1 - \rho^2) + D}{D} = \frac{1}{2} \log_2 \frac{1 - \rho^2 + D\rho^2}{D} > L_{\min}(D), \quad (40)$$

strictly, on all of $0 < D < 1$. In particular no channel attains $R_{\min}$ and $L_{\min}$ simultaneously: a clean context trades rate against content just as a noisy one does, for $D < 1 - \rho^2$ with both quantities positive, and for $D \geq 1 - \rho^2$ with $L_{\min} = 0$ but $L_R^* > 0$.

Proof. The value (39). Write $Y = \rho V + Y'$ with $Y' \perp V$, $\text{Var } Y' = 1 - \rho^2$. Converse: for any admissible channel, $I(X; \hat{Y} | V) \geq I(Y; \hat{Y} | V) \geq R_{Y|V}(D) = \frac{1}{2} \log_2^+ (\text{Var}(Y | V)/D)$, Gray's conditional function [4], exactly as in Proposition 11(ii). Achievability for $D \leq 1 - \rho^2$: with $\theta = D/(1 - \rho^2)$, take $\hat{Y} = \rho V + (1 - \theta)Y' + N$, $N \sim \mathcal{N}(0, \theta(1 - \theta)(1 - \rho^2))$: the distortion is $\theta^2(1 - \rho^2) + \theta(1 - \theta)(1 - \rho^2) = D$, and

$$I(T; \hat{Y} | V) = \frac{1}{2} \log_2 \frac{\text{Var}(\hat{Y} | V)}{\text{Var}(\hat{Y} | T)} = \frac{1}{2} \log_2 \frac{(1 - \theta)^2(1 - \rho^2) + \theta(1 - \theta)(1 - \rho^2)}{\theta(1 - \theta)(1 - \rho^2)} = \frac{1}{2} \log_2 \frac{1 - \rho^2}{D}.$$

For $D \geq 1 - \rho^2$, $\hat{Y} = \rho V$ has distortion $1 - \rho^2 \leq D$ and is V -measurable, so $I(T; \hat{Y} | V) = 0$.

The value (40). For the reverse channel, $\text{Var}(\hat{Y}_R | V) = (1 - D)^2 \text{Var}(Y | V) + \text{Var } N = (1 - D)^2(1 - \rho^2) + D(1 - D)$ and $\text{Var}(\hat{Y}_R | T) = \text{Var } N = D(1 - D)$, so

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1 - D)^2(1 - \rho^2) + D(1 - D)}{D(1 - D)} = \frac{1}{2} \log_2 \frac{(1 - D)(1 - \rho^2) + D}{D},$$

TABLE II

THE VALUES BEHIND COROLLARY 21: TWO INSTANCES WITH THE SAME REDUCED (Y, S) LAW ($\rho_{YS}^2 = \frac{1}{2}$), IN BITS. THE TWO L ROWS DIFFER AT BOTH DISTORTIONS, AND BOTH LIE BELOW THE STEINBERG ROW. EXACT FORMS FOR THE SECOND INSTANCE:

$g^* = (17 + \sqrt{229})/6$ AT $D = 0.1$ AND $(7 + \sqrt{29})/6$ AT $D = 0.3$ (DISPLAYED IN THE PROOF).

	$\text{Corr}(Y, S) = \rho/\sqrt{s}$	L at $D = 0.1$	L at $D = 0.3$
Instance $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$	$1/\sqrt{2}$	1.1610	0.3685
Instance $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$	$1/\sqrt{2}$	1.2105	0.5228
Steinberg's scalar formula on the reduced pair	$1/\sqrt{2}$	1.2297	0.5577

and $(1 - D)(1 - \rho^2) + D = 1 - \rho^2 + D\rho^2$.

Every rate-optimal channel carries the content (40). The rate moment program never references the side information, and at $\alpha = 1$ the uniqueness proof of Proposition 18 is Λ -free (the closing note there), so also at $\tau^2 = 0$ its unique minimizer is $(a, b, \nu) = (1 - D, 0, D(1 - D))$. Let a channel attain $R_{\min}$ at distortion D . The chain $R_{\min} = I(T; \hat{Y}) \geq B_0(\text{its moments}) \geq \min B_0 = R_{\min}$ forces its moments to be that triple and forces equality in (18), whose equality clause (Lemma 12) makes the conditional law of T given $\hat{Y}$ Gaussian with mean $J\hat{Y}$, $J = \mathbb{E}[T\hat{Y}]/\text{Var } \hat{Y}$, and covariance Σ_{e0} : the linear-estimation error $E := T - J\hat{Y}$ is Gaussian and independent of $\hat{Y}$. At these moments $\text{Var } \hat{Y} = (1 - D)^2 + D(1 - D) = 1 - D$ and $\mathbb{E}[Y\hat{Y}] = 1 - D$, so the Y -row of J is 1 and $Y = \hat{Y} + E_Y$ with E_Y Gaussian independent of $\hat{Y}$ and Y standard Gaussian; by Cramér's decomposition theorem [9], $\hat{Y}$ is Gaussian. Independent Gaussian vectors are jointly Gaussian, so $(T, \hat{Y}) = (J\hat{Y} + E, \hat{Y})$ is jointly Gaussian with the pinned moments: the law of the reverse channel. Its conditional content is then determined by that law,

$$I(T; \hat{Y} | V) = \frac{1}{2} \log_2 \frac{\text{Var}(\hat{Y} | V)}{\text{Var}(\hat{Y} | T)} = \frac{1}{2} \log_2 \frac{(1 - D) - (1 - D)^2 \rho^2}{D(1 - D)} = \frac{1}{2} \log_2 \frac{1 - \rho^2 + D\rho^2}{D},$$

which is (40).

Strictness. For $D < 1 - \rho^2$: $1 - \rho^2 + D\rho^2 > 1 - \rho^2$ since $D\rho^2 > 0$, so $L_R^* > L_{\min}$. For $1 - \rho^2 \leq D < 1$: $L_{\min} = 0$, while $1 - \rho^2 + D\rho^2 > D \Leftrightarrow (1 - \rho^2)(1 - D) > 0$, so $L_R^* > 0$. $\square$

Corollary 21. *Two instances whose (Y, S) joint laws are equivalent up to bijective scaling can have different minimal conditional content: the instances $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$ and $(\frac{3}{4}, \frac{1}{2})$ have $\rho_{YS}^2 = \frac{1}{2}$ in both cases, yet different L at both tested distortions; Table II lists the values, both strictly below Steinberg's scalar formula evaluated on the reduced pair. So no functional of the (Y, S) joint law, in particular neither Steinberg's scalar formula evaluated on (Y, S) nor Gray's, can equal L . Steinberg's functional is unchanged; its source here is the full pair (Y, V) , and the augmentation is essential: the encoder's access to the clean context V enters the function.*

Proof. In both instances Y is standard and $\rho_{YS}^2 = \rho^2/s = \frac{1}{2}$, and a bijective scaling of S leaves $\sigma(S)$, hence every conditional mutual information given S , unchanged, so the two instances may be normalized to a common (Y, S) law.

First instance, $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$: the context is clean. This instance sits at the boundary $\tau^2 = 0$ excluded by the standing assumption $\tau^2 > 0$ of Section III, so its value is supplied not by Theorem 14 but directly by Proposition 11(ii), which is proved at $\tau^2 = 0$ (and which the $\tau^2 \rightarrow 0$ limit of Corollary 15 matches). The instance gives

$$L = \frac{1}{2} \log_2 \frac{1}{2D},$$

so $L = \frac{1}{2} \log_2 5 = 1.1610$ bits at $D = 0.1$, and $L = \frac{1}{2} \log_2(5/3) = 0.3685$ bits at $D = 0.3$.

Second instance, $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$: $s = \frac{3}{2}$, and (20) becomes, at $D = 0.1$,

$$P(g) = \frac{3}{20}g^2 - \frac{17}{20}g + \frac{1}{4}, \quad g^* = \frac{17 + \sqrt{229}}{6} = 5.3555, \quad L = \frac{1}{2} \log_2 g^* = 1.2105 \text{ bits},$$

and, at $D = 0.3$,

$$P(g) = \frac{9}{20}g^2 - \frac{21}{20}g + \frac{1}{4}, \quad g^* = \frac{7 + \sqrt{29}}{6} = 2.0642, \quad L = \frac{1}{2} \log_2 g^* = 0.5228 \text{ bits}.$$

The value of Steinberg's scalar formula on the reduced pair: an encoder observing Y alone pays the marginalized-class optimum of Section IV, the scalar-corner value $\frac{1}{2} \log_2 [(1 - \rho_{YS}^2 + \rho_{YS}^2 D)/D]$ at $\rho_{YS}^2 = \frac{1}{2}$, that is, $\frac{1}{2} \log_2 [(1 + D)/(2D)]$: $\frac{1}{2} \log_2 5.5 = 1.2297$ bits at $D = 0.1$, and $\frac{1}{2} \log_2 (13/6) = 0.5577$ bits at $D = 0.3$. Both instances lie strictly below it ($1.1610, 1.2105 < 1.2297$ and $0.3685, 0.5228 < 0.5577$), as Proposition 11 requires. $\square$

The reason is as follows. Both instances have the same reduced correlation $\text{Corr}(Y, S) = \rho/\sqrt{s} = 1/\sqrt{2}$, realized by different splits: entirely through a clean context in the first instance ($\rho^2 = \frac{1}{2}, \tau^2 = 0$), through a stronger context read in noise in the second ($\rho^2 = \frac{3}{4}, \tau^2 = \frac{1}{2}$). The function depends on (ρ^2, τ^2) separately, not only through the ratio ρ^2/s , because the encoder's channel acts on its observation of (Y, V) , not on (Y, S) : two sources with identical (Y, S) joint laws but different internal splits of the correlation admit different optimal channels. The gap at the tabulated levels is 0.0495 bits at $D = 0.1$ and 0.1543 bits at $D = 0.3$. A pair inside the standing model shows the same: $(\rho^2, \tau^2) = (\frac{3}{5}, \frac{1}{5})$ and $(\frac{3}{4}, \frac{1}{2})$ share $\rho_{YS}^2 = \frac{1}{2}$, and for the first the closed form gives $g^* = (35 + \sqrt{745})/12$ and $(15 + \sqrt{65})/12$, so $L = 1.1880$ and 0.4712 bits at $D = 0.1$ and 0.3, against the second instance's 1.2105 and 0.5228.

Remark 22. *The closed form settles when the conditional-variance lower bound is attained. Every admissible channel obeys $L \geq I(Y; \hat{Y} | S) = h(Y | S) - h(Y | \hat{Y}, S) \geq h(Y | S) - h(Y - \hat{Y}) \geq \frac{1}{2} \log_2 (\text{Var}(Y | S)/D) = \frac{1}{2} \log_2 g_f$ with $g_f = (s - \rho^2)/(Ds)$, by the Gaussian maximum-entropy bound at variance D . Within the standing model $\tau^2 > 0$ the bound, taken as its positive part where $g_f < 1$, is attained exactly when $\rho = 0$: substituting g_f into (20) gives $P(g_f) = -\rho^2 \tau^2/s$, negative when $\rho \neq 0$, so that $g^* > g_f$ strictly; at $\rho = 0$ the quadratic factors as $(Dg - 1)(sg - 1)$ and $g_f = 1/D$ is its larger root. The clean-context case $\tau^2 = 0$, where the bound is Gray's function and is attained on $D \leq 1 - \rho^2$, is Proposition 20.*

VI. VECTOR CONTEXT

The frontier survives vector side information. Let $S = \Gamma^T X + U \in \mathbb{R}^r$, $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, with X, U , and all channel randomness mutually independent. Write $V := \Gamma^T X \in \mathbb{R}^r$ and $T := (Y, V) \in \mathbb{R}^{1+r}$, and assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping linearly dependent context components. Reproductions are scalar without loss of optimality by the data-processing reduction of Section III, which is generic in S .

Theorem 23. *Let $0 < D < 1$. (i) Sufficiency: it is without loss of optimality to let the channel depend on X only through T , and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. (ii) Whitened form: with $W, Z = WT$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{1+r})$, $0 \prec \Lambda \preceq I$, and the unit vector y_0 of (33) at $p = 1 + r$, every admissible channel with regression vector $c = \mathbb{E}[Z\hat{Y}]$ and residual variance $\nu > 0$ satisfies*

$$R \geq \frac{1}{2} \log_2 \frac{|c|^2 + \nu}{\nu}, \quad L \geq \frac{1}{2} \log_2 \frac{c^T \Lambda c + \nu}{\nu}, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + \nu, \quad (41)$$

with both bounds attained simultaneously by the jointly Gaussian channel $\hat{Y} = c^T Z + N$, $N \sim \mathcal{N}(0, \nu)$. (iii) The frontier, traced by $\alpha \in [0, 1]$: with $\gamma_0 = \nu/(|c|^2 + \nu)$ and $\gamma_1 = \nu/(c^T \Lambda c + \nu)$, the Pareto-optimal channel is the unique solution with $\nu > 0$ of the generalized two-noise-fraction system

$$c = (1 - \alpha \gamma_0 - (1 - \alpha) \gamma_1) [(1 - (1 - \alpha) \gamma_1) I + (1 - \alpha) \gamma_1 \Lambda]^{-1} y_0, \quad \nu = D - |y_0 - c|^2, \quad (42)$$

per-mode explicit given the two scalars (γ_0, γ_1) , and the frontier point is $(\frac{1}{2} \log_2(1/\gamma_0), \frac{1}{2} \log_2(1/\gamma_1))$. For $r = 1$ the program coincides with that of Theorem 14(b) under the linear bijection $c = W^{-T}(a, b)^T$, so the two systems trace the same frontier. The $\alpha = 0$ endpoint gives the vector-context function $L(D)$ as a one-dimensional root problem in γ_1 .

Proof. (i) In the resampling proof of Lemma 10, the only property of S used is that its conditional law given $(X, \hat{Y})$, and given $(X, \hat{Y}')$, is the fixed kernel $p(s | t, \hat{y}) = p_U(s - v)$, depending on x only through t . This holds verbatim for the vector $S = V + U$ with $U \sim \mathcal{N}(0, \Sigma_U)$ jointly independent of X and the channel randomness. Steps (a) through (d) of that proof are unchanged.

(ii) *Whitening.* First, $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS} \Sigma_S^{-1} \Sigma_{ST} \succ 0$, where $\Sigma_S = \Sigma_V + \Sigma_U$ is the side-information covariance (V is not renormalized in this section): positivity of $\text{Cov}(T, S)$ holds because $\Sigma_T \succ 0$ and $\Sigma_{S|T} = \text{Cov}(\Gamma^T X | T) + \Sigma_U \succeq \Sigma_U \succ 0$, and the Schur complement of a positive definite matrix is positive definite. Also $\Sigma_{T|S} \preceq \Sigma_T$.

The congruence (33) therefore exists at $p = 1 + r$ with $0 \prec \Lambda \preceq I$, and $\text{Cov } Z = I$ makes y_0 a unit vector: $Y = y_0^\top Z$, $|y_0|^2 = e_Y^\top \Sigma_T e_Y = 1$.

Moments and the two bounds. Fix an admissible channel, centered without loss, with $c = \mathbb{E}[Z\hat{Y}]$ and $\nu = \text{Var } \hat{Y} - |c|^2$; the case $\nu = 0$ is dispatched by Remark 13 (the reproduction is then a noiseless linear functional of T , so either every bound is trivial or the reproduction is null). The residual $N' = \hat{Y} - c^\top Z$ is uncorrelated with Z , hence with T and with its subvector V ; and it is uncorrelated with U by mutual independence, hence with $S = V + U$: componentwise, $\text{Cov}(N', S) = 0$. Lemma 12 (the two determinant lower bounds, with equality for the jointly Gaussian channel matching the second moments) applies with $A = c^\top W$: $A \Sigma_T A^\top = c^\top W \Sigma_T W^\top c = |c|^2$ and $A \Sigma_{T|S} A^\top = c^\top \Lambda c$, giving the two bounds of (41). The distortion evaluation is exact because Z is white: $\mathbb{E}(Y - \hat{Y})^2 = 1 - 2y_0^\top c + (|c|^2 + \nu) = |y_0 - c|^2 + \nu$. Equality for the Gaussian channel is Lemma 12's equality clause.

(iii) *The weighted program.* As in Theorem 14(b): both coordinates are convex in the channel, so the frontier is the trace of the weighted minimizations, and the weighted objective is bounded below by the moment program

$$\min_{c, \nu} \frac{\alpha}{2} \log_2 \frac{Q_0 + \nu}{\nu} + \frac{1 - \alpha}{2} \log_2 \frac{Q_1 + \nu}{\nu}, \quad Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad |y_0 - c|^2 + \nu \leq D, \quad \nu > 0,$$

whose value the Gaussian channel of (ii) attains. The objective is strictly decreasing in ν at fixed $c \neq 0$, so the distortion constraint is active at the optimum. Feasibility forces $2y_0^\top c \geq 1 - D + (|c|^2 + \nu) > 1 - D > 0$ as in (34), so $c \neq 0$ and $Q_1 \geq \lambda_{\min}|c|^2$ is bounded away from zero on the feasible set, giving coercivity as $\nu \rightarrow 0$ and attainment at an interior stationary point of the reduced objective in c , with $\nu = D - |y_0 - c|^2$.

The first-order system. Differentiating $\alpha \ln(Q_0 + \nu) + (1 - \alpha) \ln(Q_1 + \nu) - \ln \nu$ in c with $\nu = D - |y_0 - c|^2$, exactly as in the proof of Theorem 14(b) with $\nabla Q_0 = 2c$ and $\nabla Q_1 = 2\Lambda c$, and multiplying by $\nu/2$ gives (42); the bracket is invertible, each diagonal entry being a convex combination of 1 and $\lambda_j > 0$. Uniqueness of the minimizer, hence of the system's solution with $\nu > 0$, at every $\alpha \in [0, 1]$ is Proposition 18 at $p = 1 + r$, whose proof is dimension-free; achievability of the frontier point is (ii).

For $r = 1$ the bijection $c = W^{-\top}(a, b)^\top$ carries (Q_0, Q_1, h) and the constraint to their Section IV forms, so the two weighted programs are identical with the same unique minimizer, and the two stationarity systems are equivalent parameterizations of it. At $\alpha = 0$, (42) reads $c = (1 - \gamma_1)[(1 - \gamma_1)I + \gamma_1\Lambda]^{-1}y_0$; substituting into $\nu = D - |y_0 - c|^2$ and $\gamma_1 = \nu/(c^\top \Lambda c + \nu)$ leaves one scalar equation in $\gamma_1 \in (0, 1)$, the vector-context generalization of the quadratic (20). $\square$

The endpoint has a closed form after all, as an eigenvalue. Write e_Y for the coordinate vector of Y in T , so $Y = e_Y^\top T$, and

$$A_D = \Sigma_T e_Y e_Y^\top \Sigma_T - (1 - D) \Sigma_T.$$

Proposition 24. *Under the model of this section, for $0 < D < 1$,*

$$L(D) = \frac{1}{2} \log_2 \left(1 + \frac{1 - D}{\mu_{\max}} \right), \quad (43)$$

where $\mu_{\max} > 0$ is the largest root of $\det(A_D - \mu \Sigma_{T|S}) = 0$. The minimum is attained by the reproduction $\mathbb{E}[Y | W]$ of $W = b^\top T + N$, with b a generalized eigenvector for $\mu_{\max}$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, $\sigma^2 = (e_Y^\top \Sigma_T b)^2 / (1 - D) - b^\top \Sigma_T b$. For scalar context, $1 + (1 - D)/\mu_{\max} = g^*$.

Proof. By Lemma 12 with $m = 1$ and Remark 13, restrict to jointly Gaussian channels $\hat{Y} = b^\top T + N$ with $b \neq 0$ and $\text{Var } N = \sigma^2 > 0$. If $\text{Cov}(Y, \hat{Y}) = 0$ the distortion is $1 + \text{Var } \hat{Y} > D$, so $\mathbb{E}[Y | \hat{Y}]$ is a nonzero multiple of $\hat{Y}$; replacing $\hat{Y}$ by it leaves the conditional coordinate unchanged and does not raise the distortion. With $W = b^\top T + N$ reproduced by $\mathbb{E}[Y | W]$, the distortion is $1 - (e_Y^\top \Sigma_T b)^2 / (b^\top \Sigma_T b + \sigma^2)$ and the conditional coordinate is $\frac{1}{2} \log_2(1 + b^\top \Sigma_{T|S} b / \sigma^2)$, since $\text{Var}(W | S) = b^\top \Sigma_{T|S} b + \sigma^2$ and $\text{Var}(W | T) = \sigma^2$. The constraint reads $\sigma^2 \leq (e_Y^\top \Sigma_T b)^2 / (1 - D) - b^\top \Sigma_T b$, and the coordinate decreases in σ^2 , so at the largest admissible σ^2 it equals $\frac{1}{2} \log_2(1 + (1 - D) b^\top \Sigma_{T|S} b / b^\top A_D b)$, finite exactly when $b^\top A_D b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A_D b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is $\mu_{\max}$; it is positive because $b = e_Y$ gives $e_Y^\top A_D e_Y = 1 - (1 - D) = D$. For scalar context, substituting $\mu = (1 - D)/(g - 1)$ gives $\det(A_D - \mu \Sigma_{T|S}) = -(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a direct computation confirmed by computer algebra (Appendix B), so the positive root $\mu_{\max}$ corresponds to the unique root $g > 1$ of P , which is g^* . $\square$

The proposition puts the closed form of Theorem 14(a) in a different light. The quadratic $P(g)$ is the characteristic equation of a two-by-two pencil that holds the source covariance, the conditional covariance left after the side information, and the distortion level, and the achieving channel reads along its top eigenvector. That eigenvector depends on D . For a scalar described variable this is visible only as the loading $b/a = \rho/(g^*s - 1)$ on the context in (21). For a described vector it becomes a direction in the space of the described variable that turns with the distortion budget, which the companion manuscript [39] develops.

VII. THE BINARY FRONTIER

The binary case serves two purposes: it shows that the misalignment phenomenon and the strict rate–content tradeoff are not Gaussian artifacts, the frontier being fully characterized here too (Proposition 28), and it shows the functional is explicitly computable outside the Gaussian family. The conditional content itself coincides with a known common-reconstruction rate (Appendix C); the frontier is traced here, unlike its Gaussian counterpart, by an elementary self-contained argument.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V , side information $S = V \oplus W$ with $W \sim \text{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The description is the reproduction $\hat{Y} \in \{0, 1\}$ itself (a richer description cannot help: for any stored Z from which the decoder computes $\hat{Y} = f(Z)$, $I(Y, V; Z | S) \geq I(Y, V; \hat{Y} | S)$ by the data-processing inequality), and the distortion is Hamming, $\Pr[\hat{Y} \neq Y] \leq D$. Write $x * q = x(1 - q) + (1 - x)q$, $\ell(x) = \ln \frac{1-x}{x}$, and h_2 for the binary entropy function in bits.

Lemma 25. $L(C) := I(Y, V; \hat{Y} | S)$ is convex in the channel $C = P(\hat{y} | y, v)$, and the Hamming distortion is linear in C .

Proof. The Markov chain $\hat{Y} - (Y, V) - S$ means that one channel C is shared across the values of S , so $L(C) = \sum_s P(s) I_s(C)$, where $I_s(C)$ is the mutual information of the channel C under the input law $P(Y, V | S = s)$. The input laws depend on s , but each I_s is convex in C at its own fixed input law, so the nonnegative combination is convex in C . Linearity of the distortion is immediate. $\square$

Lemma 26. Fix $p \in (0, \frac{1}{2})$, $q \in (0, \frac{1}{2}]$, $D \in (0, \frac{1}{2})$. Parameterize the constraint segment $(1 - p)d_0 + pd_1 = D$, $(d_0, d_1) \in (0, 1)^2$, by $d_0 \in (\max\{0, \frac{D-p}{1-p}\}, \frac{D}{1-p})$, an open subinterval of $(0, 1)$, and define the tilt residual

$$\begin{aligned} \psi(d_0) &:= \ell(d_0) - \ell(d_1(d_0)) - 2(1 - 2q)\ell(u(d_0)), \\ d_1(d_0) &= \frac{D - (1 - p)d_0}{p}, \quad u = a * q, \quad a = 2(1 - p)d_0 + p - D. \end{aligned}$$

Then ψ has exactly one zero on the open segment, and the family value $L(d_0)$ attains its minimum over the closed segment exactly there. The same statements hold at $q = 0$ when $D < p$.

Proof. Work in nats, with h_e the binary entropy function in nats. On the symmetric family the value is $L(d_0) = h_e(u) - (1 - p)h_e(d_0) - ph_e(d_1(d_0))$, the closed form computed in the proof of Theorem 27 from the family's symmetry alone; the L of that theorem in bits is this L divided by $\ln 2$. Differentiating with $h'_e(x) = \ell(x)$, $u'(d_0) = 2(1 - p)(1 - 2q)$, and $d'_1(d_0) = -(1 - p)/p$,

$$L'(d_0) = 2(1 - p)(1 - 2q)\ell(u) - (1 - p)\ell(d_0) + (1 - p)\ell(d_1) = -(1 - p)\psi(d_0).$$

Existence and interiority. We first record the boundary behavior. At the left endpoint: if $D \leq p$ the endpoint is $d_0 = 0$ and $\ell(d_0) \rightarrow +\infty$; if additionally $D = p$ then also $d_1 \rightarrow 1$ and $-\ell(d_1) \rightarrow +\infty$, the same sign; if $D > p$ the endpoint is $d_0 = (D - p)/(1 - p) > 0$ with $d_1 \rightarrow 1$, so $-\ell(d_1) \rightarrow +\infty$ with the other terms bounded. The u term stays bounded in every case: $u = a * q \in [q, 1 - q]$ with $q > 0$. Hence $\psi \rightarrow +\infty$ at the left endpoint. At the right endpoint $d_0 = D/(1 - p) < 1$ (since $1 - p > \frac{1}{2} > D$): $d_1 \rightarrow 0$, so $-\ell(d_1) \rightarrow -\infty$ with the other terms bounded, and $\psi \rightarrow -\infty$. ψ is continuous on the open segment, so it has a zero there by the intermediate value theorem, and trivially every zero is interior.

Zeros are minimizers. The family channel is affine in (d_0, d_1) , hence affine in d_0 along the segment, and L is convex in the channel (Lemma 25), so L is convex on the segment. For a convex differentiable function, $L'(d_0) = 0$ at an interior point makes that point a global minimizer; conversely $L' \rightarrow -\infty$ at the left endpoint and $L' \rightarrow +\infty$

at the right (the sign flips of ψ), so the minimum of L over the closed segment is attained in the interior and only at zeros of ψ .

Uniqueness. The minimizer set of the convex L is a closed interval on which L is constant. Suppose it contains two distinct points. Then $L'' \equiv 0$ on a nondegenerate subinterval. But L is real analytic on the open segment: d_1 and a are affine in d_0 with values in $(0, 1)$, $u = a(1 - 2q) + q \in (0, 1)$, and h_e is real analytic on $(0, 1)$. By the identity theorem, $L'' \equiv 0$ on the whole connected open segment, so L' is constant there, contradicting $L' \rightarrow -\infty$ at the left endpoint. Hence the minimizer is unique, and since the zeros of ψ are exactly the interior stationary points of L , which are exactly its minimizers, ψ has exactly one zero.

At $q = 0$ with $D < p$ the same argument applies verbatim: the left endpoint is $d_0 = 0$ with $d_1(0) = D/p < 1$, and $u = a \in (p - D, p + D) \subset (0, 1)$ stays bounded away from 0 and 1, so the boundary blow-ups and the analyticity are as before. $\square$

Fig. 4 shows the objective, the tilt residual, and the rate–content frontier at representative instances. At $(p, q, D) = (0.25, 0.15, 0.15)$ the residual crosses zero once, at $d_0^* = 0.1173$ ($d_1^* = 0.2480$), where L attains its minimum 0.3346 bits. At $(0.1, 0.1, 0.05)$ the frontier runs from the tilt root $d_0^* = 0.0282$, where $L_{\min} = 0.4341$ bits, to $d_0 = D$, where $R_{\min} = 1 - h_2(D) = 0.7136$ bits; the L -optimal point pays 0.0391 bits of excess rate and the R -optimal point 0.0248 bits of excess conditional content.

Theorem 27. *For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} | S)$ over channels $P(\hat{y} | y, v)$ (Markov $\hat{Y} - (Y, V) - S$) with $\Pr[\hat{Y} \neq Y] \leq D$ is attained in the two-parameter symmetric family $\hat{Y} = Y \oplus E$, $\Pr[E = 1 | Y \oplus V = j] = d_j$, and equals*

$$L(D) = h_2(u) - (1 - p)h_2(d_0) - ph_2(d_1), \quad u = a * q, \quad a = (1 - p)d_0 + p(1 - d_1),$$

where, for $q \in (0, \frac{1}{2}]$ with any $D \in (0, \frac{1}{2})$, and for $q = 0$ with $D < p$, the pair (d_0, d_1) is the unique solution, interior to the constraint segment (Lemma 26), of the active constraint $(1 - p)d_0 + pd_1 = D$ together with the tilt equation, a balance of log-odds,

$$\ell(d_0) - \ell(d_1) = 2(1 - 2q)\ell(u).$$

At the boundary point $q = 0$, $D = p$, the minimum $L = 0$ is attained at the segment endpoint $(d_0, d_1) = (0, 1)$, the reproduction $\hat{Y} = V$. (In the excluded case $q = 0$, $D > p$, the constraint is inactive and $L(D) = 0$, attained non-uniquely by reproductions measurable in V alone; the tilt system still has a root there, the degenerate point $d_1 = 1 - d_0$, which evaluates to the correct value 0. For $q > 0$ the constraint is active on all of $(0, \frac{1}{2})$: L is convex, nonincreasing, positive by the last claim below, and 0 at $D = \frac{1}{2}$, hence strictly decreasing.) On the same channel the rate coordinate is $R = 1 - (1 - p)h_2(d_0) - ph_2(d_1)$ and the discount is exactly $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ (the chain-rule identity $I(Y, V; \hat{Y}) - I(Y, V; \hat{Y} | S) = I(\hat{Y}; S)$, valid under the Markov chain, in closed form: $a = \Pr[\hat{Y} \neq V]$, $u = \Pr[\hat{Y} \neq S]$). Special cases: at $q = 0$ the system gives $a = (p - D)/(1 - 2D)$ and $L = h_2(p) - h_2(D)$ (Gray's conditional rate–distortion function [4], for $D \leq p$); at $q = \frac{1}{2}$ it gives $d_0 = d_1 = D$ and $L = 1 - h_2(D)$ (the marginal rate–distortion function; the context is useless exactly when the side information is). For $q \in (0, \frac{1}{2}]$ and $D < \frac{1}{2}$, $L(D) > 0$.

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above); convexity of L in the shared channel and linearity of the distortion are Lemma 25.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1 - y, 1 - v, 1 - s, 1 - \hat{y})$ preserves the joint law of (Y, V, S) : the pair (Y, V) is doubly symmetric with fair marginals, so $(1 - Y, 1 - V) \stackrel{d}{=} (Y, V)$, and $W \sim \text{Bern}(q)$ is independent of (Y, V) , so $S = V \oplus W$ flips with V . The flip also preserves the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L . So an optimum exists among σ -symmetric channels, and these are exactly the family $\{(d_0, d_1)\}$: σ -symmetry forces $\Pr[\hat{Y} \neq Y | y, v]$ to depend only on $y \oplus v$.

Closed form. On the family, $H(\hat{Y} | Y, V) = (1 - p)h_2(d_0) + ph_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ ($J = Y \oplus V$) with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} | S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ -invariant (σ -invariant source, σ -symmetric channel), forcing equal conditional flip probabilities given either value of S ; the same symmetry gives $H(\hat{Y}) = 1$, hence R .

Tilt equation. On the active constraint segment the minimizer is the unique zero of the tilt residual, by Lemma 26.

Special cases and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which

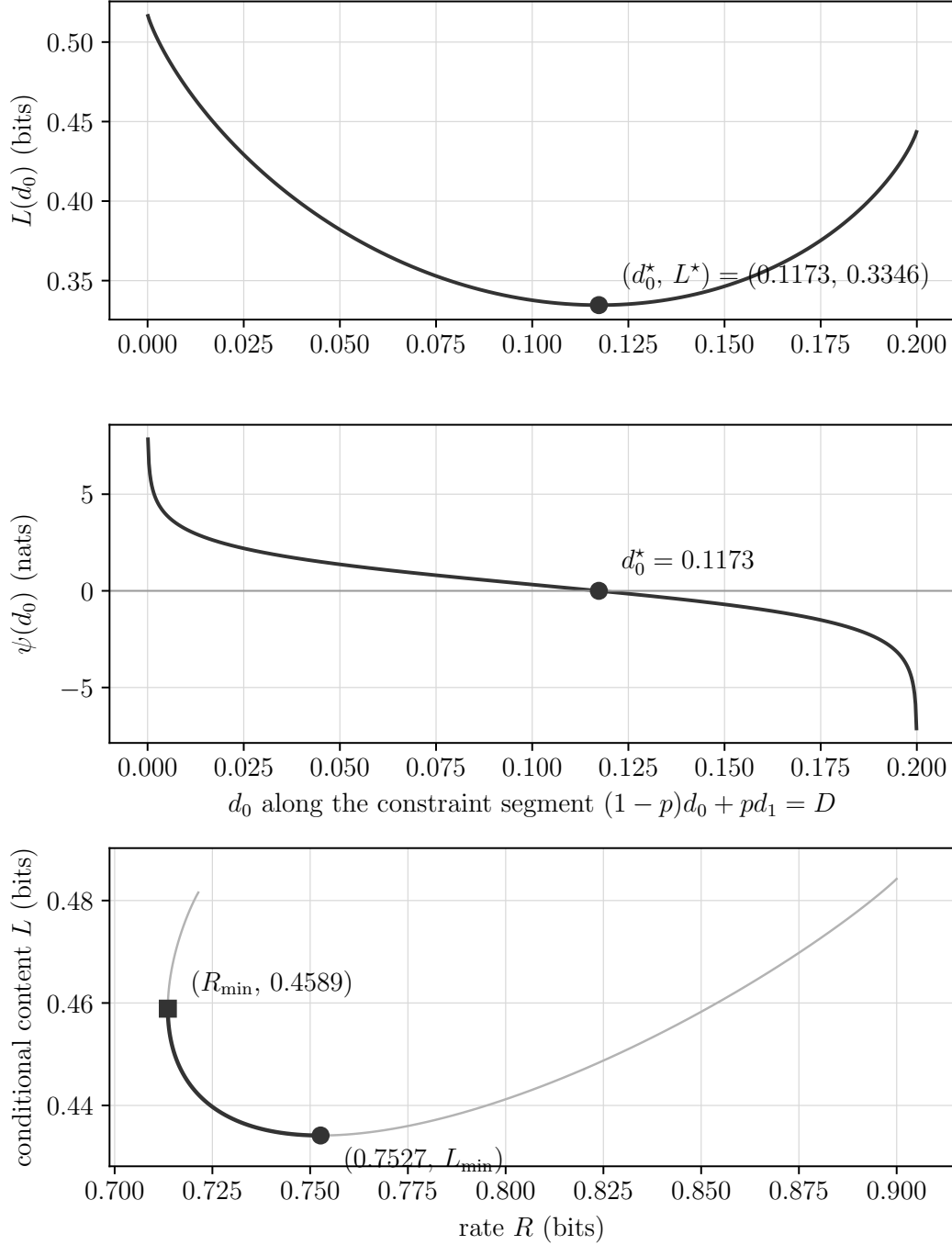


Fig. 4. The binary objective $L(d_0)$ (top) and the tilt residual $\psi(d_0)$ (middle) along the constraint segment at $(p, q, D) = (0.25, 0.15, 0.15)$, and the binary rate–content frontier (bottom) at $(p, q, D) = (0.1, 0.1, 0.05)$ with the Pareto arc from the tilt root to the rate minimizer drawn bold. The residual crosses zero once, and the arc is a strict tradeoff: each endpoint pays a positive excess in the other coordinate.

suffices by attainment): if $a \neq \frac{1}{2}$, then $h_2(u) > h_2(a) \geq (1-p)h_2(d_0) + ph_2(1-d_1) = (1-p)h_2(d_0) + ph_2(d_1)$, where the strict step holds because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$ and the second step by concavity; so $L > 0$. If $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - ph_2(d_1) \geq 1 - h_2(D) > 0$ by concavity at the distortion. $\square$

Proposition 28. *The symmetric family exhausts not only the conditional coordinate but the entire rate–content*

frontier at distortion D : the Pareto frontier of the binary rate–content region is the arc $d_0 \in [d_0^*, D]$ of the curve

$$d_0 \mapsto (R(d_0), L(d_0)), \quad R(d_0) = 1 - (1 - p) h_2(d_0) - p h_2(d_1(d_0)),$$

along the constraint segment. On the segment R is uniquely minimized at $d_0 = d_1 = D$, where $R = 1 - h_2(D)$ is the marginal rate–distortion function, and L at the tilt root d_0^* of Lemma 26. For $q \in (0, \frac{1}{2})$ the two minimizers are distinct, $d_0^* < D$, and $[d_0^*, D]$ is a strict tradeoff arc, L increasing and R decreasing, with both coordinates worsening outside it; at $q = \frac{1}{2}$ the frontier is a single corner.

Proof. The rate coordinate $R = I(Y, V; \hat{Y})$ is convex in the shared channel by the argument of Lemma 25 with S deleted, so every weighted objective $\alpha R + (1 - \alpha)L$, $\alpha \in [0, 1]$, is convex in the channel. The global bit flip σ of Theorem 27’s proof preserves the joint law of (Y, V, S) , the distortion, and both coordinates: the flip is a law-preserving bijective relabeling of both arguments of each mutual information, so $R(C^\sigma) = R(C)$ and $L(C^\sigma) = L(C)$. Averaging a channel with its flip therefore preserves the distortion and, by convexity, does not increase any weighted objective, and larger reproduction alphabets are covered by the data-processing reduction in both coordinates. Every weighted optimum is thus attained in the family $\{(d_0, d_1)\}$, and the distortion constraint may be taken active there: both coordinates are nonnegative and vanish at $d_0 = d_1 = \frac{1}{2}$, so by convexity the objective does not increase along the straight path from any point with distortion slack toward $(\frac{1}{2}, \frac{1}{2})$, while the distortion, affine in (d_0, d_1) with positive weights, rises continuously to $\frac{1}{2} > D$ and meets the budget on the way. Hence the frontier lies on the displayed curve along the segment, and the arc between the two minimizers identified next is the Pareto-optimal part. On the segment, R is strictly convex with its unique minimum at $d_0 = d_1 = D$ (strict concavity of $(1 - p)h_2(d_0) + p h_2(d_1)$ and the stationarity $\ell(d_0) = \ell(d_1)$), where $R = 1 - h_2(D)$, the marginal rate–distortion function; L has its unique minimum at the tilt root d_0^* (Lemma 26). For $q \in (0, \frac{1}{2})$ the two minimizers are distinct, since $\psi(D) = -2(1 - 2q)\ell(u(D)) < 0$ (there $u(D) < \frac{1}{2}$ because $a(D) = p + D(1 - 2p) < \frac{1}{2}$), so $d_0^* < D$: on $[d_0^*, D]$ the curve is a strict tradeoff arc, L increasing and R decreasing, and outside that arc both coordinates worsen. At $q = \frac{1}{2}$ the two minimizers coincide at $d_0 = D$ and the frontier is a single corner. Fig. 4 traces the frontier at a representative instance. $\square$

The conditional-content function of this section is the common-reconstruction rate of Lu *et al.* [23] under the relabeling of Appendix C; the contribution here is the joint (R, L) frontier. The binary common-reconstruction calculations of Steinberg [12], with value $h_2(\rho_0 * D) - h_2(D)$ at side-channel crossover ρ_0 , and of Ahmadi, Tandon, Simeone, and Poor [14], for erased side information, place S at the decoder and let the encoder observe the source alone. Here S is held by a third party that never decodes and the encoder observes the pair (Y, V) . An encoder restricted to Y alone faces the constrained problem of the pair (Y, S) , a binary symmetric pair with crossover $p * q$, and pays Steinberg’s value there (the marginalization identity of Section IV is alphabet-free); the tilt equation is what access to the pair adds.

VIII. DISCUSSION AND CONCLUSION

This paper characterized the minimal conditional content of a description whose encoder observes the pair (Y, V) while the side information S stays with a third party: the closed form of Theorem 14(a), the exact rate–content region of Theorem 14(b) with the misalignment condition of Corollary 19, the pencil characterization of Proposition 24 with the vector-context extension of Theorem 23, and the binary frontier of Theorem 27 and Proposition 28. The structural conclusion is Corollary 21: the function is not determined by the reduced marginal (Y, S) but by what the encoder can see, so conditional description content, the rate at which a party holding the side information can re-encode the description, is a property of the encoding configuration, not of the source and side-information pair alone. Stated generally, the paper establishes in an exactly solvable case that the cost of a description is relative to the party that pays it. One description carries one rate but as many conditional contents as there are parties holding different side information, and the encoder’s observations, not the source alone, decide how those costs trade off. The closed form and the region make the principle quantitative, and the open problems below are its natural next instances.

Two problems are open beyond the vector described variable, which the companion manuscript [39] treats. Stationary sources, and the process and spectral versions of the function, are not treated here. The finite-blocklength behavior of the pair (R_n, L_n) is unexplored, and the content coordinate is not a routine second-order corollary of

the rate one: the dispersion of L_n is governed by the variance of a conditional information density, which the quantize-and-pass-to-the-limit argument of Appendix A does not control, so a joint (R_n, L_n) dispersion region does not follow from the rate dispersion alone.

The closed forms were checked numerically against direct minimization (Appendix B).

APPENDIX A THE GAUSSIAN OPERATIONAL THEOREM

This appendix proves the operational region theorem for the jointly Gaussian source of Section III without appeal to an abstract-alphabet extension of the finite-alphabet theorem. The source letter is the pair $T = (Y, V)$ of Section III, jointly Gaussian, as Definition 2 fixes it; a code whose encoder observes the ambient source of Section III is covered as well, since any T^n -code is such a code, and for any ambient channel the resampled T -channel of Lemma 10 keeps the distortion and the conditional coordinate and has rate $I(T; \hat{Y}') \leq I(X; \hat{Y})$, so the union over ambient channels equals the union over T -channels. The side information is $S \in \mathbb{R}^r$ as in Section III ($r = 1$) or Section VI; the distortion is $d(x, \hat{y}) = (y - \hat{y})^2$, unbounded; codes, R_n , D_n , and $L_n = H(M | S^n)/n$ are as in Section II, and achievability is Definition 1. The proof gives the converse by the chain rule, restricts to Gaussian channels, proves closedness, and proves achievability by quantizing the side information (Step A), then the source and the reproduction (Step B), and then returning to the unquantized side information (Step C).

One device is fixed first, the *saturating dyadic quantizers*: for $\kappa \geq 1$, q_κ partitions $\mathbb{R}$ into the bounded cells $[j2^{-\kappa}, (j+1)2^{-\kappa})$, $-4^\kappa \leq j < 4^\kappa$, and the two saturating tail cells $(-\infty, -2^\kappa)$ and $[2^\kappa, \infty)$, and maps each point of a bounded cell to the cell's left endpoint and the two tail cells to -2^κ and 2^κ , so that $|q_\kappa(u) - u| \leq 2^{-\kappa} + |u| \mathbf{1}\{|u| > 2^\kappa\}$ for every real u . A vector is quantized coordinatewise, so its cells are rectangles. The partitions are nested, every cell edge of q_κ being an edge of $q_{\kappa+1}$, so for any random vector Z the σ -fields $\sigma(q_\kappa(Z))$ increase with κ , and their union generates $\sigma(Z)$, since the dyadic rectangles generate the Borel σ -field. Write $S_\kappa := q_\kappa(S)$.

Theorem 29. *For the Gaussian source under the standing assumptions of Section III ($\tau^2 > 0$, $\Sigma_T \succ 0$), with the pair $T = (Y, V)$ as the source letter X (Definition 2), and every $0 < D < 1$,*

$$\mathcal{RW}(D) = \bigcup_{p(\hat{y}|x): \mathbb{E}(Y - \hat{Y})^2 \leq D} \{(R, L) : R \geq I(X; \hat{Y}), L \geq I(X; \hat{Y} | S)\},$$

and the union is unchanged when restricted to jointly Gaussian test channels.

The compactness step of the proof uses $D < 1$. For $D \geq 1$ both sides equal $[0, \infty)^2$: the zero reproduction meets the distortion constraint with $R = L = 0$, and $R_n, L_n \geq 0$ for every code. For $D = 0$ both sides are empty as subsets of $\mathbb{R}^2$: a code with finitely many messages has $D_n > 0$ and $R_n \geq \frac{1}{2} \log_2(1/D_n)$, so no finite pair is achievable at distortion 0, and the only channels with zero distortion have $\hat{Y} = Y$ almost surely, with both coordinates infinite. Throughout, L_n is the coding quantity (6); no step of the proof depends on Remark 8.

Proof. Converse. The chain of Theorem 6's converse (the discrete operational region theorem) holds for the continuous source. M is discrete, so $\log |\mathcal{M}_n| \geq H(M) \geq I(X^n; M) \geq I(X^n; \hat{Y}^n)$ and $H(M | S^n) \geq I(X^n; M | S^n) \geq I(X^n; \hat{Y}^n | S^n)$ are unchanged. The single-letterization needs no differential entropy. Since X_i is independent of X^{i-1} , the chain rule gives $I(X^n; \hat{Y}^n) = \sum_i I(X_i; \hat{Y}^n, X^{i-1}) \geq \sum_i I(X_i; \hat{Y}_i)$; and since the pair (X_i, S_i) is independent of the other pairs, $I(X_i; X^{i-1}, S^n) = I(X_i; S_i)$, so $I(X^n; \hat{Y}^n | S^n) = \sum_i [I(X_i; \hat{Y}^n, X^{i-1}, S^n) - I(X_i; S_i)] \geq \sum_i [I(X_i; \hat{Y}_i, S_i) - I(X_i; S_i)] = \sum_i I(X_i; \hat{Y}_i | S_i)$. Every term is finite: each is at most $I(X_i; S_i) + H(M)$, where $I(X_i; S_i) = \frac{1}{2} \log_2(s/\tau^2)$ is finite because $\tau^2 > 0$ and $H(M) \leq \log |\mathcal{M}_n|$ because $\hat{Y}^n$ is a function of the finite-valued M ; this is a second place where $\tau^2 > 0$ is used. The per-letter induced channels satisfy the Markov constraint by the i.i.d. structure, exactly as in the finite case; the mixture channel $\bar{q}$ is defined by averaging the induced conditional laws, its distortion is D_n , and the convexity step is Lemma 3 in the general-alphabet form of Remark 4, which holds for the conditional coordinate because, under $\hat{Y} - X - S$, $I(X; \hat{Y} | S)$ is the integral over s against P_S of the relative entropy between the joint law $P_{X|S=s} q$ of $(X, \hat{Y})$ given $S = s$ and the product $P_{X|S=s} \times (P_{X|S=s} q)$ of its marginals; both arguments are affine in q , so the integrand is convex in q by the joint convexity of relative entropy, and so is the integral. Hence (R_n, L_n) lies in the displayed union at distortion D_n .

Restriction to Gaussian channels. For any admissible channel the hypotheses of Lemma 12 are met after one normalization. Centering $\hat{Y}$ leaves both coordinates and the Markov chain unchanged and lowers the distortion

by $(\mathbb{E}\hat{Y})^2$; $\mathbb{E}\hat{Y}^2 < \infty$ follows from $\mathbb{E}(Y - \hat{Y})^2 \leq D$ and $\mathbb{E}Y^2 = 1$; the residual of the regression of $\hat{Y}$ on T is uncorrelated with S by the argument in the proof of Theorem 14(a), since U is independent of $(X, \hat{Y})$; and a positive residual variance may be assumed, because a zero residual with nonzero regression coefficients gives both coordinates infinite and an empty quadrant (Remark 13), while a zero residual with zero coefficients is the zero reproduction, infeasible for $D < 1$. With these in place, Lemma 12 (Gaussian exhaustion: the two determinant lower bounds, met with equality by the jointly Gaussian channel matching the second moments) produces the jointly Gaussian channel with the same second moments and the same distortion, whose coordinates $I(X; \hat{Y})$ and $I(X; \hat{Y} | S)$ are componentwise no larger than the original channel's. Its quadrant therefore contains the original channel's quadrant, so the union over all channels equals the union over jointly Gaussian channels. It thus suffices to prove achievability for a jointly Gaussian test channel.

Closedness and the passage from D_n to D . By the restriction step, the union at any distortion level $D' \in (0, 1)$ equals the union over jointly Gaussian test channels, and these are parameterized by their regression vector and residual variance: in the whitened coordinates of (33) (the simultaneous congruence $\Sigma_T \rightarrow I$, $\Sigma_{T|S} \rightarrow \Lambda$), by pairs $\theta = (c, \nu)$ with $Q_0 = |c|^2$, $Q_1 = c^\top \Lambda c$, distortion $|y_0 - c|^2 + \nu \leq D'$, and coordinates $(\frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu}, \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu})$. The distortion constraint confines θ to a compact set: $|y_0 - c|^2 \leq D'$ is a closed ball and $0 \leq \nu \leq D'$; and feasibility forces $c \neq 0$ there, since $c = 0$ gives distortion at least $|y_0|^2 = 1 > D'$. Both coordinates are lower semicontinuous on that set with values in $(0, \infty]$, finite and continuous where $\nu > 0$ and diverging as $\nu \rightarrow 0$ at $c \neq 0$. Now let (R_j, L_j) lie in the union at levels D'_j with $\limsup_j D'_j \leq D$ and $(R_j, L_j) \rightarrow (R, L)$ finite. Pick feasible θ_j whose quadrant contains (R_j, L_j) ; the θ_j eventually lie in the compact set at level $D + \delta$ for each $\delta \in (0, 1 - D)$, so a subsequence converges to some θ^* , feasible at D by continuity of the distortion in θ . Lower semicontinuity gives $\frac{1}{2} \log_2 [(Q_0 + \nu)/\nu](\theta^*) \leq \liminf_j \frac{1}{2} \log_2 [(Q_0 + \nu)/\nu](\theta_j) \leq R$, and likewise for the conditional coordinate, so (R, L) lies in θ^* 's quadrant, hence in the union at D . This establishes both closedness at fixed distortion and right-continuity in the distortion level, and with $\limsup D_n \leq D$ places every limit point of (R_n, L_n) in the union at D .

Achievability, Step A: quantizing the side information. Fix a jointly Gaussian admissible channel $p(\hat{y} | x)$ with distortion at most D and residual variance $\nu > 0$ (a channel with $\nu = 0$ has both coordinates infinite by the coordinate formulas of the closedness step, the conditional one because $\Lambda \succ 0$ when $\tau^2 > 0$, and contributes nothing to the union), and fix $\epsilon > 0$. Along the saturating dyadic quantizers,

$$I(\hat{Y}; S_\kappa) \uparrow I(\hat{Y}; S) \quad (\kappa \rightarrow \infty) : \quad (44)$$

the sequence is nondecreasing because the partitions are nested (data processing), and its limit is $I(\hat{Y}; S)$ because the σ -fields $\sigma(S_\kappa)$ increase to $\sigma(S)$ and mutual information is continuous along an increasing sequence of σ -fields whose union generates the limit [6, Ch. 2]. Choose κ with

$$I(\hat{Y}; S_\Delta) \geq I(\hat{Y}; S) - \epsilon, \quad S_\Delta := S_\kappa. \quad (45)$$

S_Δ is a function of S , so the Markov chain $\hat{Y} - X - S_\Delta$ holds, and by the chain-rule identity used throughout $(I(X; \hat{Y}) - I(X; \hat{Y} | S')) = I(\hat{Y}; S')$ for any side information S' with $\hat{Y} - X - S'$,

$$I(X; \hat{Y} | S_\Delta) = I(X; \hat{Y}) - I(\hat{Y}; S_\Delta) \leq I(X; \hat{Y} | S) + \epsilon. \quad (46)$$

Achievability, Step B: discretizing the source and the reproduction. The pair source is now (X, S_Δ) with continuous X and finite S_Δ , and the test channel is the jointly Gaussian $p(\hat{y} | x)$. Theorem 6 requires finite alphabets and a bounded distortion, so we discretize both and pass to the limit. For $m \geq 1$ let $X_m := q_m(X)$, the coordinatewise quantizer applied to $X = (Y, V)$, and $\hat{Y}_m := q_m(\hat{Y})$, so that $|\hat{Y}_m - \hat{Y}| \leq 2^{-m} + |\hat{Y}| \mathbf{1}\{|\hat{Y}| > 2^m\}$ and $\hat{Y}_m \rightarrow \hat{Y}$ in L^2 , since $\mathbb{E}\hat{Y}^2 < \infty$ by the distortion constraint and $\mathbb{E}[\hat{Y}^2 \mathbf{1}\{|\hat{Y}| > 2^m\}] \rightarrow 0$ by dominated convergence. The finite-alphabet surrogate consists of the source X_m and the side information S_Δ under their true joint law, which is that of a discrete memoryless pair because (X_m^n, S_Δ^n) is a coordinatewise function of the i.i.d. pairs (X^n, S^n) ; the finite reproduction alphabet $q_m(\mathbb{R}) \subset \mathbb{R}$; the test channel

$$p_m(\hat{y} | x_m) := \Pr(\hat{Y}_m = \hat{y} | X_m = x_m),$$

the conditional law of the quantized reproduction given the quantized source; and the surrogate distortion

$$d_m(x_m, \hat{y}) := \mathbb{E}[(Y - \hat{y})^2 | X_m = x_m] = \text{Var}(Y | X_m = x_m) + (\mathbb{E}[Y | X_m = x_m] - \hat{y})^2,$$

which is bounded, since both alphabets are finite. Both conditional expectations are well defined, because $\Sigma_T \succ 0$ gives every rectangle of q_m positive probability under the law of X . Under the surrogate joint law $p(x_m, s_\Delta) p_m(\hat{y} | x_m)$ the Markov chain $\hat{Y}_m - X_m - S_\Delta$ holds by construction, so Lemma 7 applies to the surrogate and, for every $\epsilon > 0$ and every m , supplies a deterministic code, a function of X_m^n alone, with $|\mathcal{M}_n| \leq 2^{n(I(X_m; \hat{Y}_m) + 2\epsilon)}$, surrogate distortion at most $\mathbb{E}_m d_m + \epsilon + 2\epsilon_n d_{\max}(m)$, and

$$H(M | S_\Delta^n) \leq n(I_m(X_m; \hat{Y}_m | S_\Delta) + 3\epsilon) + 1 + 2\epsilon_n n(I(X_m; \hat{Y}_m) + 2\epsilon), \quad (47)$$

where $\mathbb{E}_m$ and I_m are computed under the surrogate joint law, $d_{\max}(m) = \max_{x_m, \hat{y}} d_m(x_m, \hat{y})$ is finite for each m but grows with m because the tail cells sit at $\pm 2^m$, and $\epsilon_n \rightarrow 0$ for each fixed (ϵ, κ, m) . The limits below are taken in this order: ϵ is fixed; κ is fixed by Step A; m is fixed by (ii) and (iii); then $n \rightarrow \infty$, so that $2\epsilon_n d_{\max}(m) = o(1)$ and $2\epsilon_n n(\cdot) = o(n)$, which is how they are written below; and finally $\epsilon \downarrow 0$ along a diagonal sequence. Also, the surrogate law of the pair $(X_m, \hat{Y}_m)$ is their true joint law, so $I(X_m; \hat{Y}_m)$ needs no subscript. Three facts transfer these guarantees to the original problem.

(i) *The code is a code for the original source, with the same distortion.* X_m^n is a function of X^n , so the code is admissible for X^n . Each reproduction letter $\hat{y}_i$ is a function of X_m^n , and given X_m^n the conditional law of Y_i depends on $X_{m,i}$ alone, by memorylessness; hence $\mathbb{E}[(Y_i - \hat{y}_i)^2 | X_m^n] = d_m(X_{m,i}, \hat{y}_i)$, and taking expectations and averaging over i identifies the true distortion with the surrogate distortion.

(ii) *The surrogate distortion converges to the original one.* Since the surrogate law of $(X_m, \hat{Y}_m)$ is the true one,

$$\mathbb{E}_m d_m(X_m, \hat{Y}_m) = \mathbb{E}[\text{Var}(Y | X_m)] + \mathbb{E}[(\mathbb{E}[Y | X_m] - \hat{Y}_m)^2] \rightarrow \mathbb{E}(Y - \hat{Y})^2 \leq D \quad (m \rightarrow \infty),$$

because the nested partitions generate $\sigma(X)$, so $\mathbb{E}[Y | X_m] \rightarrow Y$ in L^2 by the martingale convergence theorem, which sends the first term to zero, and $\hat{Y}_m \rightarrow \hat{Y}$ in L^2 . (The surrogate distortion is not $\mathbb{E}(Y - \hat{Y}_m)^2$ under the true law, since Y and $\hat{Y}_m$ are dependent given X_m ; the argument does not use that identity.)

(iii) *The surrogate coordinates approach the original ones.* The rate obeys $I(X_m; \hat{Y}_m) \leq I(X; \hat{Y})$ by data processing. For the conditional coordinate, the chain-rule identity under the surrogate Markov chain gives $I_m(X_m; \hat{Y}_m | S_\Delta) = I(X_m; \hat{Y}_m) - I_m(\hat{Y}_m; S_\Delta) \leq I(X; \hat{Y}) - I_m(\hat{Y}_m; S_\Delta)$, so it remains to show that $\liminf_m I_m(\hat{Y}_m; S_\Delta) \geq I(\hat{Y}; S_\Delta)$. The surrogate law of $(\hat{Y}_m, S_\Delta)$ is $Q_m(B \times \{j\}) = \mathbb{E}[\Pr(\hat{Y}_m \in B | X_m) \Pr(S_\Delta = j | X_m)]$, and for every bounded function f on $\mathbb{R} \times q_\kappa(\mathbb{R}^r)$, continuous in its first argument,

$$\begin{aligned} \int f dQ_m &= \mathbb{E} \left[\sum_j \mathbb{E}[f(\hat{Y}_m, j) | X_m] \Pr(S_\Delta = j | X_m) \right] \\ &\rightarrow \mathbb{E} \left[\sum_j \mathbb{E}[f(\hat{Y}, j) | X] \Pr(S_\Delta = j | X) \right] = \mathbb{E}f(\hat{Y}, S_\Delta) : \end{aligned}$$

for each j , $\mathbb{E}[f(\hat{Y}_m, j) - f(\hat{Y}, j)] \rightarrow 0$ by bounded convergence, since $\hat{Y}_m \rightarrow \hat{Y}$ pointwise and f is bounded and continuous in its first argument, and $\mathbb{E}[f(\hat{Y}, j) | X_m] \rightarrow \mathbb{E}[f(\hat{Y}, j) | X]$ in L^1 by the martingale convergence theorem, so $\mathbb{E}[f(\hat{Y}_m, j) | X_m] \rightarrow \mathbb{E}[f(\hat{Y}, j) | X]$ in L^1 by the triangle inequality; the factors $\Pr(S_\Delta = j | X_m)$ are bounded by one and converge in L^1 to $\Pr(S_\Delta = j | X)$ by the same theorem, so each product converges in L^1 and the finite sum passes to the limit; and the last equality is the conditional independence of $\hat{Y}$ and S_Δ given X , the Markov chain $\hat{Y} - X - S$. So Q_m converges weakly to the true law of $(\hat{Y}, S_\Delta)$, and mutual information is lower semicontinuous under weak convergence of the joint law [8]. Hence for all large m , $I_m(\hat{Y}_m; S_\Delta) \geq I(\hat{Y}; S_\Delta) - \epsilon$, and with (46),

$$I_m(X_m; \hat{Y}_m | S_\Delta) \leq I(X; \hat{Y}) - I(\hat{Y}; S_\Delta) + \epsilon = I(X; \hat{Y} | S_\Delta) + \epsilon \leq I(X; \hat{Y} | S) + 2\epsilon.$$

Fix m so large that (ii) gives $\mathbb{E}_m d_m \leq D + \epsilon$ and (iii) holds. The selected code then has rate at most $I(X; \hat{Y}) + 2\epsilon$, true distortion at most $D + 2\epsilon + o(1)$ by (i), and, by (47) and (iii),

$$H(M | S_\Delta^n) \leq n(I(X; \hat{Y} | S) + 5\epsilon + o(1)). \quad (48)$$

Achievability, Step C: back to the unquantized side information. S_Δ^n is a function of S^n , so $\sigma(S_\Delta^n) \subseteq \sigma(S^n)$ and conditioning on the finer σ -algebra cannot increase entropy:

$$H(M | S^n) \leq H(M | S_\Delta^n) \leq n(I(X; \hat{Y} | S) + 5\epsilon + o(1)), \quad (49)$$

by (48).

Collect. For every $\epsilon > 0$ the triple $(I(X; \hat{Y}) + 2\epsilon, I(X; \hat{Y} \mid S) + 5\epsilon, D + 2\epsilon)$ is achievable; a diagonal sequence $\epsilon_j \downarrow 0$ of codes (the ϵ_j -code exists at every sufficiently large blocklength by Lemma 7, and blocklengths $N_j \leq n < N_{j+1}$ use it) achieves, in the limsup sense of Definition 1, the corner $(I(X; \hat{Y}), I(X; \hat{Y} \mid S))$ at distortion D , and the quadrant above the corner is immediate from the definition. Together with the converse and the restriction step, this proves the theorem. $\square$

The two imported results are the monotone convergence of mutual information under refining partitions [6, Ch. 2] and its lower semicontinuity under weak convergence [8]. Every other step is Theorem 6 or is proved here; no abstract-alphabet covering or Markov lemma is invoked.

APPENDIX B NUMERICAL CHECKS

The closed forms and the displayed matrix identities were checked independently of the derivations: each closed form numerically, against direct minimization over test-channel moments, and each identity symbolically, by computer algebra. For Proposition 24, `verify_pencil.py` confirms the scalar identity symbolically and compares the pencil formula with direct minimization for random vector contexts. The check scripts and an executed notebook, `reproduce.ipynb`, which evaluates the closed forms against direct minimization and regenerates the figures and tables, accompany the manuscript sources at the repository of the title footnote and are submitted as supplementary material. The checks guard against algebraic slips; no proof in the paper depends on them.

APPENDIX C

COMPARISON WITH THE COMMON-RECONSTRUCTION FUNCTIONS OF LU *et al.* AND OF LU AND XU

Lu, Xu, Zhang, Feng, and Wang [23], with the Heegard–Berger companion [24], solve the binary problem in which the encoder observes a source X and side information S_1 , the decoder holds S_2 with $X - S_1 - S_2$, and the encoder must reproduce the decoder’s estimate. For the doubly symmetric source with $\Pr[X \neq S_1] = p_1$ and $S_2 = S_1 \oplus \text{Bern}(p_2)$ under Hamming distortion, their Theorem 1 gives the rate–distortion function

$$\tilde{R}^{\text{CR}}(D) = \min_{W: W - (X, S_1) - S_2, \mathbb{E}d(X, f(W)) \leq D} I(X, S_1; W \mid S_2),$$

reduced to a one-parameter minimization in their Theorem 2. Under the relabeling $X \leftrightarrow Y$, $S_1 \leftrightarrow V$, $S_2 \leftrightarrow S$, $W \leftrightarrow \hat{Y}$, $(p_1, p_2) \leftrightarrow (p, q)$, their functional is the conditional-content functional of Section VII under the same Markov chain and constraint, so the binary $L(D)$ coincides with $\tilde{R}^{\text{CR}}(D)$ and we do not claim it as new; the endpoints agree, Gray’s $h_2(p) - h_2(D)$ at $q = 0$ (their Remark 3), $1 - h_2(D)$ at $q = \frac{1}{2}$, and $h_2(p * q)$ at $D = 0$ (their Remark 1). In their problem the quantity is a transmitted rate; here it is a second cost beside the unconditional rate $R = I(Y, V; \hat{Y})$, which their problem does not constrain, so the joint region and the frontier of Proposition 28 are the binary contribution of this paper.

Lu and Xu’s Gaussian function

Lu and Xu’s vector Gaussian Heegard–Berger problem with common reconstructions [25] is the closest Gaussian counterpart. Their Theorem 4 gives the rate–distortion function for two decoders with side information under error-covariance constraints $\text{Cov}(X - \hat{X}) \preceq \mathbf{D}_m$, with $\mathbf{D}_m$ diagonal and positive definite, under the hypothesis $\mathbf{D}_m \preceq \Sigma_X$ in the positive semidefinite order. Specialized to one decoder with side information S and constraint $\text{diag}(D, D_V)$ on the pair (Y, V) , with $s = 1 + \tau^2$, it reads $\frac{1}{2} \log_2[(1 - \rho^2)(\tau^2 + D_V)/(sDD_V)]$: their Assumption 1 holds in the (Y, V) coordinates, since $\Sigma_{T|S}^{-1} - \Sigma_T^{-1} = \text{diag}(0, \tau^{-2})$, and the same specialization returns Steinberg’s scalar formula when the context is dropped. The present problem is the unpenalized-context limit $D_V \rightarrow \infty$ of the underlying Lu–Xu constraint family, at which the covariance constraint reduces to $\mathbb{E}(Y - \hat{Y})^2 \leq D$; every channel feasible for their constraint is feasible here, and dropping $\hat{V}$ lowers the conditional information by data processing, so their function is at least $L(D)$ wherever it is defined. What their closed form does not cover is that limit: its hypothesis $\mathbf{D}_m \preceq \Sigma_X$ forces $(1 - D)(1 - D_V) \geq \rho^2$, which leaves no admissible D_V unless $D < 1 - \rho^2$ and never leaves the context unpenalized. On the admissible range the inequality is strict: the specialized value decreases in D_V , and at the largest admissible D_V it equals $\frac{1}{2} \log_2 g_R$ with $P(g_R) = \rho^2 \tau^4 (1 - D)(1 - \rho^2)/(s(1 - D - \rho^2)^2) > 0$,

so $g_R \neq g^*$ while $g_R \geq g^*$ by the inequality just proved. The formal continuation of the formula to $D_V = \text{Var } V$ equals Gray's clean-context function $\frac{1}{2} \log_2((1 - \rho^2)/D)$, strictly below $L(D)$ whenever $\rho\tau \neq 0$, falling further beyond. Theorem 14(a) is therefore the value of a limit that their closed form is not stated for and does not continue to.

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